WeedReaver: A Precision Weed Management System for Punjab Wheat and Cotton

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Students' Submission

We certify that this report is submitted as the FYP-1 mid-report for the project titled *WeedReaver: A Precision Weed Management System for Punjab Wheat and Cotton*.

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Date of Submission: 11 October 2026

Anti-Plagiarism Statement

The undersigned students hereby certify that this report is our own original work completed under the supervision of Dr. Labiba Fahad at the FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad. All sources, literature citations, software libraries, datasets, and benchmark references used in preparing this report have been appropriately cited and acknowledged in accordance with university academic integrity regulations and IEEE citation standards.

We further declare that no part of this report has been plagiarised or fabricated, and the similarity index strictly adheres to HEC and FAST-NUCES anti-plagiarism guidelines (Turnitin similarity below 19% with no individual source exceeding 5%). All technical figures, architectural diagrams, schemas, and data protocol specifications have been authored directly by the project team.

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Dated: 11 October 2026

Authors' Declaration

We declare that the report describes the project team's work and distinguishes repository evidence from demonstration behavior and planned research. Individual contribution claims must be checked against team records before this declaration is signed.

- **Asad Amad (22I-1329):** Individual repository contributions should be stated from verified author, pull-request, and team records.
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- **Abdul Rafay Ashfaq (22I-0954):** Individual repository contributions should be stated from verified author, pull-request, and team records.

The audited repository uses a FastAPI backend, React/TypeScript dashboard, Kotlin/Jetpack Compose Android source, PostgreSQL/SQLite persistence, and a configurable NodeODM adapter. Its segmentation defaults are simulated; a trained model and its field evaluation are not claimed as completed work.

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Abstract

WeedReaver is a survey-guided system for recording weed patches and planning targeted treatment in Punjab wheat fields. The current research focus is wheat, with little-seed canary grass (*Phalaris minor*) as the principal target weed because it competes with wheat and can be difficult to distinguish from the crop at early growth stages. The system must distinguish a target weed from the crop, other vegetation, and uncertain observations; it does not claim to detect every weed species. Cotton (*Gossypium hirsutum*) remains a separate future extension. Its Kharif calendar and weed community require local cotton data and a separate evaluation before any cotton performance claim.

The repository contains a FastAPI backend, a React/TypeScript dashboard, and Kotlin/Jetpack Compose Android client source for field workflows. OpenDroneMap integration is configurable, but default photogrammetry and aerial-segmentation engines are simulated; the model-backed segmentation adapter is not implemented. Android screens include demo scan outcomes, and no on-device weed model asset is present. Demonstration records are synthetic and wheat-focused. Therefore, this report distinguishes implemented interfaces and demo behavior from field data collection and model validation.

The proposed contribution is a crop-bounded, spatially explicit workflow that connects survey records, a human-reviewable weed map, treatment-zone records, and follow-up verification. Its empirical value remains to be established using locally collected wheat imagery, agronomist-checked labels, spatially separated evaluation data, and field observations. A later cotton extension will require its own season, labels, and evaluation. No yield gain, herbicide saving, local model accuracy, or cotton result is claimed as an achieved result in this report.

Keywords: Precision Agriculture, Wheat, Weed Mapping, Phalaris minor, Geospatial Analysis, OpenDroneMap, Human Review, Cotton Extension.

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We extend our heartfelt thanks to the faculty and administration of the **FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad**, for providing an intellectually rigorous environment and the computational infrastructure required to conduct this work.

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Executive Summary

Problem and Scope

WeedReaver is a survey-guided weed-mapping and treatment-record platform. The first study is limited to Punjab Rabi wheat and candidate patches of *Phalaris minor*, a competitive grass weed that can resemble wheat in early growth. It does not claim to detect every weed. The annotation plan separates crop, confirmed target weeds, other vegetation, and uncertain observations; treatment decisions require agronomic review.

Cotton (*Gossypium hirsutum*) remains a future extension, not a validated domain in the current repository. Its Kharif timing and weed community differ from wheat, so local cotton imagery, labels, and evaluation are needed before cotton performance or advice can be claimed. Candidate cotton weeds in the literature include broadleaf, grass, and sedge species, but those are not implemented target labels in the current wheat demo [1].

Repository Status

The GitHub repository contains a FastAPI backend, a React/TypeScript dashboard, and Kotlin/Jetpack Compose Android client source. These provide field and season records, map/grid/zone analysis, survey and review workflows, treatment and verification records, exports, and related APIs. A NodeODM adapter is available when configured. The default processing and segmentation engines are simulated; the model-backed segmentation adapter is explicitly unimplemented. Android scan results are predefined demo outcomes, and no on-device weed model asset is present. The README's statement that the field app is separate conflicts with Android source in the checkout.

User Workflow

An analyst signs in, selects a field and season, reviews the survey and candidate weed map, adjusts the map threshold, inspects zones and uncertain items, and records treatment or

follow-up. The dashboard represents this analyst workflow. Android screens provide a coded operator prototype, but their scan outcomes are demo values. The panel-requested clickable Figma prototype and user evaluation remain outstanding.

Research Contribution and Data

The proposed contribution is an interactive geospatial workflow that links field and survey records to reviewable candidate zones and follow-up data. Its agronomic value remains untested. The next phase needs real wheat imagery, documented flight and crop-stage metadata, agronomist-reviewed labels, field-separated evaluation, and user feedback. Synthetic demo data are excluded from model training and accuracy claims. Cotton, if pursued, will use a separate Kharif data protocol and evaluation.

Work Remaining

1. Confirm field access, consent, and flight permissions; measure processing requirements with a real image set and configured NodeODM host.

2. Collect and annotate wheat imagery centred on wheat and *P. minor*, retaining non-target and uncertain categories.

3. Implement a model-backed aerial segmentation engine and evaluate it on fields held out from training.

4. Evaluate the analyst dashboard with intended users and validate the separate field-operator journey independently.

5. Measure treatment outcomes only with a documented field comparison and observed data. Do not report demo values as efficacy or savings.

Defense Feedback

This revision narrows the first study, defines the primary weed target and label boundaries, explains what data are synthetic, clarifies the user workflow, and separates implemented code from future work. The report does not claim that panel comments about presentation duration, delivery, or the Figma prototype have been completed; those require presentation and prototype work outside this report.

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List of Abbreviations

Abbreviation	Complete Expansion
AGL	Above Ground Level
ALS	Acetolactate Synthase (Herbicide Group 2 / B)
ACCase	Acetyl-CoA Carboxylase (Herbicide Group 1 / A)
API	Application Programming Interface
ArUco	Augmented Reality University of Cordoba (Square Binary Fiducial Marker)
CWFID	Crop / Weed Field Image Dataset
DAS	Days After Sowing
DFD	Data Flow Diagram
EPSG	European Petroleum Survey Group (Geodetic Parameter Registry)
ERD	Entity-Relationship Diagram
FP32	32-bit Floating Point Precision
FR	Functional Requirement
GIS	Geographic Information System
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GSD	Ground Sampling Distance
HRAC	Herbicide Resistance Action Committee
INT8	8-bit Signed Integer Quantization
JSON	JavaScript Object Notation
JWT	JSON Web Token
mIoU	Mean Intersection over Union
NARC	National Agricultural Research Centre (Islamabad)
NFR	Non-Functional Requirement
NNAPI	Android Neural Networks API
ODM	OpenDroneMap (Photogrammetric Reconstruction Engine)
PCAA	Pakistan Civil Aviation Authority
PIDE	Pakistan Institute of Development Economics
POC	Proof of Concept
PVC	Polyvinyl Chloride

Abbreviation	Complete Expansion
RAM	Random Access Memory
REST	Representational State Transfer
RGB	Red, Green, Blue (Visible Spectrum Color Model)
SoC	System on Chip
SRS	Software Requirements Specification
UAV	Unmanned Aerial Vehicle
UI	User Interface
UML	Unified Modeling Language
UTM	Universal Transverse Mercator
WGS84	World Geodetic System 1984 (EPSG:4326)
XNNPACK	Highly Optimized Neural Network Inference Operators Library
YOLO	You Only Look Once
Zadoks	Standard Cereal Plant Growth Stage Scale (Z10–Z90)

Chapter 1

Introduction and Project Vision

1.1 Project Overview

WeedReaver is a survey-guided weed-mapping and treatment-record platform. The first research study is limited to Rabi wheat in Punjab, with little-seed canary grass (*Phalaris minor*) as the primary target weed. This grass weed competes with wheat and can be difficult to distinguish from the crop at early growth stages. The system is not intended to detect every weed or to treat every green plant.

Cotton (*Gossypium hirsutum*) remains a future extension in this report. Cotton is a Kharif crop with different field structure, weed species, and management timing. The repository is wheat-focused and does not provide a locally validated cotton model or complete cotton-specific workflow. Cotton data and results must therefore remain separate from the wheat study.

The checked repository contains a FastAPI backend and an interactive React/TypeScript analyst dashboard. Field, survey, map, review, treatment, and verification workflows are present. However, demonstration field surfaces are synthetic, simulated segmentation is the default, and the model-backed aerial segmentation adapter is not implemented. The separate Android field application described in the repository README is not included in the audited repository. This report distinguishes these existing software components from planned field collection and evaluation.

1.2 Problem Statement

A field-wide view can hide where weeds remain after treatment, while walking every area to find patches takes time. For the first study, the bounded problem is: *Can a georeferenced wheat survey and an interactive map help an analyst identify and review candidate*

Phalaris minor patches for targeted follow-up, using labels and field observations that can be checked independently? The study will first test whether the data and workflow are reliable; it will not assume a model can identify all weeds or that a map alone proves that spraying is effective.

1.3 Target Weed and Label Policy

Phalaris minor is the principal target because it is a competitive wheat weed and is identified in the repository's wheat demonstration catalog. Other example labels in that catalog include *Avena ludoviciana*, *Chenopodium album*, and *Convolvulus arvensis*. These are demonstration labels, not evidence of reliable field detection. An agronomist must confirm the final target list and labels for any collected data.

The annotation policy distinguishes wheat crop, confirmed target weeds, other or non-target vegetation, and uncertain/background observations. Plants outside the agreed target list are not automatically treated as harmful weeds. Uncertain cases are held for human review, and treatment decisions remain subject to agronomic advice and product labels. Cotton candidate weeds such as *Trianthema* spp., *Cynodon dactylon*, and *Cyperus rotundus* are recorded as literature-informed examples only; they are not implemented or validated targets [1].

1.4 Project Vision

The intended workflow is concise: an analyst selects a wheat field and season, reviews a survey and candidate weed map, checks uncertain observations, adjusts and reviews treatment zones, records treatment, and later compares follow-up observations. The current web dashboard supports the analyst-facing screens and map interaction. Real image-derived predictions, a separate field-operator client, and field efficacy are future validation work.

1.5 Project Objectives

- **O1:** Maintain traceable wheat field, season, survey, and treatment records in the backend.

- **O2:** Evaluate whether the geospatial grid and zone calculations present candidate patches consistently in the dashboard.

- **O3:** Collect a wheat dataset with recorded provenance and agronomist-reviewed labels for wheat, *P. minor*, other vegetation, and uncertain cases.
- **O4:** Implement an aerial model only after suitable data and labels are available; evaluate it on fields excluded from training and report class-wise errors.
- **O5:** Evaluate the analyst workflow with intended users and specify the separate field-operator workflow without presenting it as an implemented repository component.
- **O6:** Measure treatment response only through a documented field protocol and follow-up observations. Treat cotton as a separate future extension requiring local Kharif data and independent evaluation.

1.6 Project Scope

1.6.1 In-Scope for the Current Study

- **Crop and region:** Rabi wheat in Punjab is the first implementation and evaluation domain.
- **Primary target:** *Phalaris minor*; other species remain candidate labels until confirmed by an agronomist and supported by enough data.
- **Software baseline:** FastAPI backend, React/TypeScript dashboard, geospatial grid/zone analysis, survey and treatment records, and configurable NodeODM integration.
- **Data:** A small local, traceable wheat dataset with field, date, growth-stage, capture, label, and review metadata. Current synthetic demonstration data are excluded from training and accuracy evaluation.
- **Cotton:** Retained as a separately scoped extension. No cotton performance or treatment recommendation is claimed from wheat data.

1.6.2 Out of Scope at This Checkpoint

- Detecting all weed species or making unattended chemical treatment decisions.
- Claiming model accuracy, herbicide savings, efficacy, or yield improvement from synthetic demonstration surfaces.
- Claiming a completed on-device model or field-operator mobile application based on this repository.

- Transferring wheat labels, model outputs, treatment intervals, or chemical guidance directly to cotton.

1.6.3 Data and Evaluation Boundary

The seeded station and infestation surfaces in the repository are synthetic demo content. They are suitable for exercising screens and business logic only. The next data phase must collect real wheat imagery and ground observations, document label provenance, and separate training and evaluation by field or spatial block. Any cotton study requires a distinct Kharif capture plan, local labels, and crop-specific evaluation.

1.7 Project Modules

The repository's current implementation is organized around backend, dashboard, and processing components. The module names below describe the report's workflow boundaries; they should not be read as proof that every planned field or mobile capability is complete.

1.7.1 Core Modules

1. **Module C1: Field and Season Records:** Backend records field boundaries, crop seasons, surveys, and related operational data. The current demonstration seed and product catalog are wheat-focused.

2. **Module C2: Image Processing and Map Display:** Backend supports flight-image handling and a configurable NodeODM adapter. The simulated default does not produce an orthomosaic; real processing depends on a configured NodeODM service.

3. **Module C3: Segmentation and Zone Analysis:** Grid and zone calculations are implemented for stored or simulated surfaces. The default aerial segmentation is synthetic and the model-backed adapter is not implemented.

4. **Module C4: Review and Verification Records:** The backend and dashboard support review and verification records. A field-operator mobile client and on-device inference are outside the audited repository.

5. **Module C5: Prescription Review:** The dashboard presents a threshold-driven map and treatment zones. Offline navigation by a knapsack applicator is a planned integration.

6. **Module C6: Treatment and Follow-Up Records:** The software stores treatment and verification information. Follow-up change values from the demo are synthetic and are not field efficacy measurements.

1.7.2 Deferred Modules and Schedule Buffers

Field deployment, a real aerial model, a separate offline field application, user evaluation, cotton support, and agronomic efficacy trials remain planned work. They should be scheduled only after the wheat data protocol, field access, and model evaluation plan are confirmed.

1.8 Stakeholders and User Classes

The system serves four primary stakeholder classes, detailed in Table 1.1.

Table 1.1: Stakeholder Personas, Operational Surfaces, and Literacy Profiles

Role	Operational Surface	Core System Interactions	Technical Profile
Farm Manager / Owner	Desktop Web Dashboard	Defines field boundaries, inspects aerial infestation maps, reviews prescription costs, and tracks historical HRAC chemical rotation logs.	Moderate computer literacy; access to desktop and broadband.
UAV Survey Pilot	Ground Controller / Laptop	Executes pre-programmed autonomous flight plans, logs sortie metadata, and uploads raw capture frames to the processing pipeline.	High technical proficiency in drone operations and GIS.
Knapsack Field Operator	Planned field client	Would review published zones and record work; this client is separate from the audited repository and requires user testing.	Requires clear, field-usable instructions.
Agronomist / Extension Officer	Review collaborator	Confirms target species and annotation decisions and helps interpret field observations.	Agronomic expertise is required for label review.

1.9 Success Criteria

The following are evaluation criteria for future work, not achieved results. Numerical thresholds will be set after data quality, field access, and the evaluation protocol are established.

Table 1.2: Success Criteria and Evidence Required

Criterion	Technical Specification	Evidence
Map calculation	Grid and zone outputs agree with fixed reference fixtures and remain interpretable in the dashboard.	Reproducible software test output.
Wheat weed identification	Report class-wise performance for wheat and the agreed target weed, with field-separated evaluation.	Labeled local imagery and held-out field results.
User workflow	Intended users can review a map, inspect uncertain cases, and record a treatment decision.	Observed task completion and documented feedback.
Treatment outcome	Compare follow-up observations against a defined field protocol and suitable comparison.	Field measurements; no estimate from synthetic demo data.

1.10 Constraints and Assumptions

1.10.1 System Constraints

- **Photogrammetric Resources:** Real NodeODM processing needs suitable compute, storage, and image data. The repository does not establish a hardware memory limit; measure resource use on the selected host before fixing flight-size requirements.
- **Seasonal Agronomic Capture Window:** Wheat capture is planned during its Rabi vegetative window (December–February). Cotton requires a separate Kharif capture window, site access, and crop-stage schedule; full-season cotton efficacy and yield validation are outside the current FYP-2 calendar unless the project schedule is extended.
- **Field Client Boundary:** The separate Android client and any on-device model need independent implementation and device validation.

1.10.2 Explicit Engineering Assumptions

To ensure the project fails loudly rather than silently during development, five foundational assumptions are explicitly documented alongside their empirical testing protocols in Table 1.3.

Table 1.3: Controlled Project Assumptions and Falsification Protocols

Assumption	Validation Protocol	Contingency if Falsified
1. Candidate patches are useful to review	Compare mapped candidate zones with georeferenced field observations on held-out wheat plots.	Revise the map workflow or target if agreement is poor.
2. Field access and consent	Confirm site access, landholder consent, and safe flight conditions before collecting imagery.	Reschedule or use another consenting site; do not claim field results.
3. UAV regulatory feasibility	Check current applicable rules and obtain required permissions before any flight.	Defer the flight or use another approved capture method.
4. Map location accuracy	Compare the georeferenced map and operator location against field reference points.	Adjust map resolution and workflow to measured uncertainty.
5. Open-source photogrammetry viability	Configure NodeODM and record outcome, image metadata, processing time, and resource use.	Reduce image batch size or revise capture settings if processing fails.

1.11 Ethical, Legal and DEI Considerations

1.11.1 Airspace and UAV Regulatory Compliance

Applicable UAV rules and permissions must be checked with the relevant authorities before any field flight. The audited repository does not document permission to fly or confirm a field site. The study plan therefore treats authorization and landholder consent as prerequisites:

1. Obtain written site access and landholder consent before collecting field imagery.
2. Confirm airspace, operator, equipment, and flight requirements for the selected site and date.

3. Use visual-line-of-sight and other applicable safety procedures established for the approved flight.

4. If flight approval is unavailable, revise the data collection method and document its limitations; no camera-pole fallback is claimed as built or validated.

1.11.2 Chemical Advisory and Legal Liability Bounds

A student software platform should not replace approved product labels or agronomic advice. The following are design boundaries for the proposed workflow, not a claim that every interface currently enforces them:

- Do not use the system to calculate chemical application rates, volume per acre, or spray pressure.
- Any future advisory must be crop-specific and checked against current local product labels and extension guidance. Wheat examples must not be presented as cotton recommendations.
- User-facing screens should direct operators to approved product labels and qualified local advice for dose and protective equipment.

1.11.3 Data Privacy, Farmer Consent, and Accessibility (DEI)

Field coordinates, imagery, and farmer property records require consent, access control, and a documented retention plan before collection or release. The report does not claim that a public dataset has been anonymized or released.

The Android source contains operator-oriented screens, but no field usability study is documented. Text, color, icons, and touch targets should be reviewed with intended users and adapted to local language and literacy needs before operational use.

Chapter 2

Research and Existing Solutions

2.1 Research Method and Approach

The background study reviews weed-management context, aerial and ground sensing, image datasets, and spatial evaluation methods. It informs the proposed wheat workflow but does not replace local field observations or an agronomist-approved label set:

1. **Crop-Specific Agronomic Evidence Review:** Wheat literature was reviewed for *Phalaris minor*, resistance, and cereal growth stages; cotton sources were reviewed only to define a possible later extension [1, 3, 7, 8, 9, 10, 11, 12, 18]. Wheat findings are not used as cotton thresholds [19, 20].

2. **Public Precision Agriculture Benchmark Audit:** The team reviewed PhenoBench [21], WeedMap [22], WeedsGalore [23], CWFID [24], CottonWeedDet12 [2, 25], CoFly-WeedDB [26], and CamelinaWeed [27], comparing annotation types, imaging conditions, and crop domains. CottonWeedDet12 is a cotton-specific benchmark from the southern United States; its transfer to Punjab is an experiment, not evidence of local performance.

3. **Cross-Crop Evaluation Design:** Wheat and cotton results will be reported in separate crop-specific splits. Public cotton imagery can test external domain shift and pipeline compatibility, but cannot establish local Punjab cotton accuracy because it differs in geography, species mix, and capture conditions. Local cotton evaluation requires field imagery and agronomist-verified labels from the Kharif season [1, 2].

4. **Architectural and Algorithmic Exploration:** The team investigated deep convolutional and vision transformer architectures for semantic segmentation [28, 29, 30], graph-based crop-row representation models [31, 32, 33, 34], selective classification theory [35], and on-device neural quantization runtime frameworks (LiteRT / TensorFlow Lite with XNNPACK and NNAPI hardware delegates).

2.2 Existing Solutions and Prior Art

2.2.1 Commercial Industrial Solutions

In the global commercial domain, precision weed management has been pioneered through high-cost, machinery-mounted computer vision systems:

- **John Deere See & Spray (Ultimate / Premium):** Tractor-boom-mounted cameras operating at 1.5 m height capturing high-frame-rate RGB imagery, executing deep learning inference to actuate high-frequency solenoid nozzles. While representing the state of the art in spot-spraying, the system requires massive capital expenditure ($100,000–$250,000), relies on high-clearance 120-foot booms incompatible with small, bunded Punjab fields, and requires continuous subscription fees.

- **Bilberry / Trimble WeedSeeker 2 / Weed-IT:** Optical sensor systems utilizing active near-infrared (NIR) and red reflectance to detect chlorophyll fluorescence (NDVI). While robust for "green-on-brown" fallow weed control, these sensors struggle with "green-on-green" discrimination of weeds nestled inside a living, green crop canopy, and cost over $4,000 per meter of boom width.

- **Greeneye Technology:** Dual-camera boom systems combining broadleaf detection in corn and soybean with spot-spraying nozzles. Similar to See & Spray, the platform is restricted to wide-row row crops (75 cm spacing) and requires heavy agricultural tractors.

- **DJI Terra and DJI Agras Ecosystem:** Commercial drone mapping software that stitches aerial multispectral or RGB imagery and exports variable-rate prescription shapefiles for automated agricultural spraying drones (Agras T30/T40). While technically capable of aerial mapping, DJI Terra operates as a closed, proprietary, cloud-tethered desktop suite; critically, it lacks dedicated green-on-green deep learning models for grass weed discrimination in wheat, does not support manual knapsack applicators, and provides no multi-temporal chemical resistance tracking.

2.2.2 Academic Prior Art and Benchmarks

In the academic literature, vision-based weed management has evolved across distinct paradigms:

- **CWFID (Crop / Weed Field Image Dataset):** An early benchmark featuring 40 ground-level RGB images of carrots and broadleaf weeds [24]. Due to its tiny

volume, controlled lighting, and ground-level perspective, it offers no utility for aerial cereal mapping.

- **WeedMap:** A large-scale multispectral and RGB UAV dataset collected over sugar beet fields at 10 m to 20 m AGL [22]. While pioneering aerial semantic segmentation, WeedMap addresses wide-row dicot crops (sugar beet) where plants are geometrically isolated, an assumption that fails completely in narrow-row drill cereals.

- **PhenoBench:** A premier high-resolution benchmark capturing sugar beet at 1.5 m AGL via a ground-based mobile phenotyping robot [21]. Annotations provide pixel-level semantic, instance, and leaf-level masks. However, PhenoBench represents ground-level robotics rather than aerial photogrammetry, focuses on sugar beet, and lacks multi-temporal chemical treatment tracking.

- **RoWeeder and Graph Convolutional Networks (GCNs):** Recent unsupervised and semi-supervised frameworks that construct spatial graphs representing individual plants, exploiting the rigid geometric regularity of crop planting rows to detect intra-row and inter-row weeds [31, 33, 34].

2.3 Comparison of Existing Solutions

Table 2.1 summarizes selected system categories discussed in the cited sources. Cost, crop coverage, and treatment-loop capabilities vary by product and deployment; the table is a high-level comparison, not a controlled benchmark.

2.4 Limitations and Gaps in Existing Literature

The literature and repository review identify issues that the wheat study must address:

2.4.1 1. The Prohibitive Capital Cost Barrier

Some commercial precision-spraying systems require specialized machinery and may be difficult to access for small farms. The proposed workflow uses a web map and existing field records, but its equipment and operating costs have not yet been estimated for the target users.

Table 2.1: High-Level Comparison of Selected Weed-Management Approaches

System / Platform	Sensing Tier	Green-on-Green in Wheat	Hardware Cost Barrier	Offline Edge Mobile	Treatment Verification Loop
John Deere See & Spray	Tractor Boom (1.5 m)	Yes (Corn/Soy; limited wheat)	Extremely High ($150k+)	No (Proprietary Cab)	No (Single-pass actuation)
Weed-IT / WeedSeeker 2	Tractor Boom (Active NIR)	Poor (Primarily Green-on-Brown)	High ($40k–$80k)	No	No
DJI Terra / Agras	Drone Aerial (30–50 m)	No (Vegetation index threshold)	Moderate ($15k–$25k)	No (Cloud / Desktop)	No (Unverified spray pass)
PhenoBench / Robotics	Ground Rig (< 2 m)	Yes (Sugar beet only)	High (Research robot)	No	No (Static dataset)
RoWeeder / Graph CNNs	Ground / Low UAV	Limited (Breaks in narrow rows)	Low–Moderate	No (Desktop GPU)	No
WeedReaver (in development)	**UAV data path + web dashboard + Android prototype**	**Planned; not model-validated**	**Not measured**	**Local storage/sync code present; field evaluation pending**	**Records and demo workflow present; field verification pending**

2.4.2 2. The Failure of the "Crop-Row Prior" in Drill Wheat

In some row-crop settings, including cotton, plants are established in more widely separated rows than narrow-drilled wheat. Row geometry can provide useful spatial context, but it does not guarantee that all inter-row vegetation is a weed or that all crop plants remain on a perfect line; planting methods and canopy development vary. Any cotton row prior must therefore be tested on local data, while the wheat analysis below addresses its own narrow-drill geometry [1, 31, 34].

Row geometry may provide context, but it is not a reliable universal classifier. Wheat planting pattern, canopy stage, and weed distribution should be recorded and evaluated before using row cues:

1. **Narrow Row Geometry:** Wheat is sown using seed drills with extremely narrow row spacing (15 cm to 22 cm). At a flight altitude of 15 m AGL (GSD $\approx$ 0.35 cm/pixel), the inter-row gap occupies only 40 to 60 pixels.
2. **Rapid Canopy Overlap and Closure:** By the start of tillering (Zadoks Z22–Z25), wheat leaf blades extend 10 cm to 15 cm laterally, causing adjacent rows to interlace and completely close the visual inter-row canopy, occluding soil lines.
3. **Criss-Cross and Broadcast Sowing Realities:** In many Punjab districts, wheat is planted via double-pass cross-drilling or tractor-drawn broadcast cultivator passes (*chhatta*), creating pseudo-random plant distributions with zero persistent linear row structure.

For the first wheat study, the model should not depend solely on an assumed regular row pattern. Whether morphology or other cues can distinguish *P. minor* from wheat remains an empirical question.

2.4.3 3. Spatial Autocorrelation and Data Leakage in Vision Benchmarks

A methodological risk in aerial image evaluation is spatial leakage. Nearby or overlapping tiles can share plants, lighting, soil, and canopy patterns. A random tile split may therefore overstate performance on new fields:

- A single continuous weed patch or crop canopy is partitioned across overlapping tile boundaries, placing half of the plant in the training fold and the other half in the test fold.
- Tiles from nearby parts of one field may be too similar across training and test partitions.

- Performance on a held-out field may differ from a random within-field split; the size of any difference must be measured for the dataset used.

The wheat study will compare field- or spatial-block evaluation with a random-tile baseline and report the split method. It will not assume a particular amount of metric inflation.

2.4.4 4. Complete Absence of a Closed Management and Verification Loop

Many weed-detection studies focus on image-level or pixel-level model performance. WeedReaver's proposed contribution is to connect map review with treatment and follow-up records in one software workflow. The extent to which that integration helps operators or improves treatment decisions remains to be tested.

2.5 The Proposed Solution: WeedReaver

The current repository provides a web dashboard, backend records, geospatial analysis, and Android prototype screens. The following are proposed capabilities to evaluate, not completed field results:

1. **Geospatial review:** Present candidate map zones for analyst review using the current dashboard.

2. **Wheat-focused model:** Implement an aerial segmentation model only after the local data and label protocol are ready.

3. **Human review:** Route ambiguous or out-of-scope observations to an agronomist or trained operator rather than forcing a species claim.

4. **Spatial evaluation:** Test field- or block-separated evaluation and document the result rather than promising leakage-free generalization.

5. **Treatment records:** Use follow-up records to support a later field study; demo values do not measure treatment efficacy.

2.6 Technical Challenges and Research Questions

The proposed wheat study raises practical questions that the current demo cannot answer:

1. **Image and label quality:** Determine whether the available RGB imagery and field observations can distinguish wheat from candidate *Phalaris minor* patches at the selected growth stages.
2. **Spatial evaluation:** Compare a random tile split with a field- or spatial-block split; report the observed difference for this dataset rather than assuming a fixed leakage effect.
3. **Map usefulness:** Measure agreement between candidate zones and ground observations, and test whether the dashboard presents enough context for a human reviewer.
4. **Processing resources:** Measure NodeODM time, memory, and storage for the selected image set and host before setting flight-batch limits.
5. **Treatment follow-up:** Establish a crop-appropriate observation protocol before estimating treatment response. The default simulated follow-up is not a biological measurement.

2.7 Research Considerations and Design Decisions

Table 2.2 summarizes the research considerations guiding the next evaluation. These are questions and design choices, not measured findings from the current repository.

Table 2.2: Research considerations and corresponding evaluation choices

Research consideration	**Evaluation choice**
Spatially adjacent aerial tiles may share image content and field conditions.	Compare field- or spatial-block evaluation with a random-tile baseline and report the observed difference.
Wheat and cotton differ in crop structure, season, and weed community.	Limit the first study to wheat; keep any later cotton data and results separate.
Aerial imagery may not resolve species identity at every location.	Separate crop, confirmed target weeds, other vegetation, and uncertain cases; route uncertain cases to human review.
A dashboard map does not itself establish that a treatment decision is useful.	Evaluate map-review tasks with intended users and compare candidate zones with field observations.
Synthetic demo surfaces can look like real treatment results.	Mark simulation data clearly and exclude it from model accuracy and agronomic efficacy calculations.

Chapter 3

Project Plan and Defense Modifications

3.1 Proposal Defense Response

The panel asked the team to narrow the problem, avoid broad detection claims, explain target weeds and data, make the user workflow visible, clarify resources, strengthen the research contribution, reduce slide text, improve diagrams, and add clickable functionality to the Figma prototype. The revised report addresses scope, data boundaries, repository status, and the dashboard workflow. Presentation delivery and the Figma prototype remain separate tasks.

3.2 Iteration 1: Repository Baseline

3.3 Remaining Work

1. Confirm wheat field access, consent, safe flight permissions, and available equipment. Candidate research sites are not treated as confirmed until written access is obtained.

2. Finalize the target label guide with an agronomist. Separate wheat, *Phalaris minor*, other vegetation, and uncertain observations.

3. Collect a small traceable wheat image set with field, date, stage, camera, and annotation metadata. Partition evaluation by held-out field or spatial block.

4. Implement the real segmentation model and record software tests, model metrics, failure cases, and the exact data split.

Table 3.1: Panel recommendations and current response

Recommendation	**Response**	**Status**
Narrow the research problem	First evaluation is Punjab Rabi wheat and candidate *Phalaris minor* patches.	Report updated
Avoid unrealistic all-weed claims	State the label boundary and human-review handling for other/uncertain vegetation.	Report updated
Clarify data and contribution	Synthetic demo data are separated from the planned local data; contribution is stated as a testable workflow, not a proven agronomic result.	Data/model pending
Demonstrate front-end workflow	Describe analyst flow in the existing dashboard and Android source; distinguish demo screens from field performance.	Report updated
Clickable Figma prototype	Keep this as a separate design deliverable.	Pending
Improve slides and delivery	Reduce slide text and rehearse within the allocated time.	Pending outside report
Include cotton	Keep cotton as a future extension with independent Kharif data and evaluation.	Scope clarified

Table 3.2: Evidence-based status of the current repository

Area	**Repository evidence**	**Boundary**
Backend	FastAPI API, domain models, database migrations, operational routes, and tests.	Test run evidence should be attached separately.
Dashboard	React/TypeScript web application with analyst pages and interactive weed-map controls.	User study not evidenced.
Android client	Kotlin/Compose source, Room persistence, synchronization, and field workflow screens.	Demo scan outcomes; no model asset. README and source tree disagree about whether this client is part of the repo.
Photogrammetry	NodeODM adapter and simulated default engine.	Real processing requires service, imagery, and resource measurement.
Aerial segmentation	Synthetic default engine and model adapter interface.	Model implementation and field performance pending.
Field data	Seeded wheat demo data and synthetic infestation/follow-up values.	No training or validation data.

5. Reconcile the README/client-source mismatch, demonstrate dashboard and Android screen flows, and build the clickable Figma prototype requested by the panel.

6. Pursue cotton only when a Kharif site, crop-stage plan, local weed labels, and sufficient evaluation time are available.

3.4 Responsibilities

Table 3.3: Responsibilities to assign and document for the next iteration

Workstream	**Responsibility to confirm with the team**
Field access and capture	Site permissions, UAV/operator availability, field metadata, and safety documentation.
Agronomy and labels	Weed target confirmation, annotation guide, label review, and field observations.
Model and data evaluation	Training implementation, field-separated split, metrics, and error analysis.
Web, Android, and prototype	Dashboard workflow, Android demo boundary, API integration, and clickable Figma navigation.

3.5 Milestones

Table 3.4: Next milestones, subject to access and schedule feasibility

Milestone	**Exit evidence**	**Status**
M2.1 Wheat data protocol	Approved label guide, site access, permissions, and capture metadata plan.	Pending
M2.2 Wheat pilot data	Documented imagery, field observations, and annotation review.	Pending
M2.3 Model baseline	Versioned model, held-out field evaluation, class-wise metrics, and failure examples.	Pending
M2.4 User workflow	Dashboard and Android tasks observed with intended users; clickable Figma prototype demonstrated.	Pending
M2.5 Agronomic follow-up	Protocol and observed comparison data supporting any efficacy or savings claim.	Pending

3.6 Risk Controls

The first risk controls are to avoid flying before permissions and consent, avoid model training before labels and splits are defined, mark simulated data clearly in demos, and prevent any result from being generalized from wheat to cotton without separate evidence. Processing RAM, image-batch limits, handset constraints, and field usability should be measured with the selected hardware instead of asserted in advance.

Chapter 4

Software Requirements

4.1 User Interactions and Use Cases

This use-case view summarizes the intended user workflow and distinguishes the current web/Android prototypes from future field validation:

- **Farm Manager / Landowner:** Selects a field and season, reviews candidate weed-map zones, and checks treatment and follow-up records. Savings and agronomic effect are not established by demo data.
- **UAV Survey Pilot:** Uploads survey imagery and metadata. Real orthomosaic generation requires a configured NodeODM service; the default engine is simulated.
- **Field Knapsack Applicator:** Uses the Android prototype's map, route, treatment, and verification screens. Offline persistence/sync code is present, but field navigation and usability have not been validated.
- **Agronomist / Extension Specialist:** Confirms weed labels and reviews uncertain observations. Model-based abstention and species predictions remain future work.
- **OpenDroneMap (ODM) Processing Engine:** External service used by the configurable real-image processing adapter.

4.2 Use Case Diagram

Figure 4.1 shows current and proposed user tasks; it does not imply that every task is field-validated.

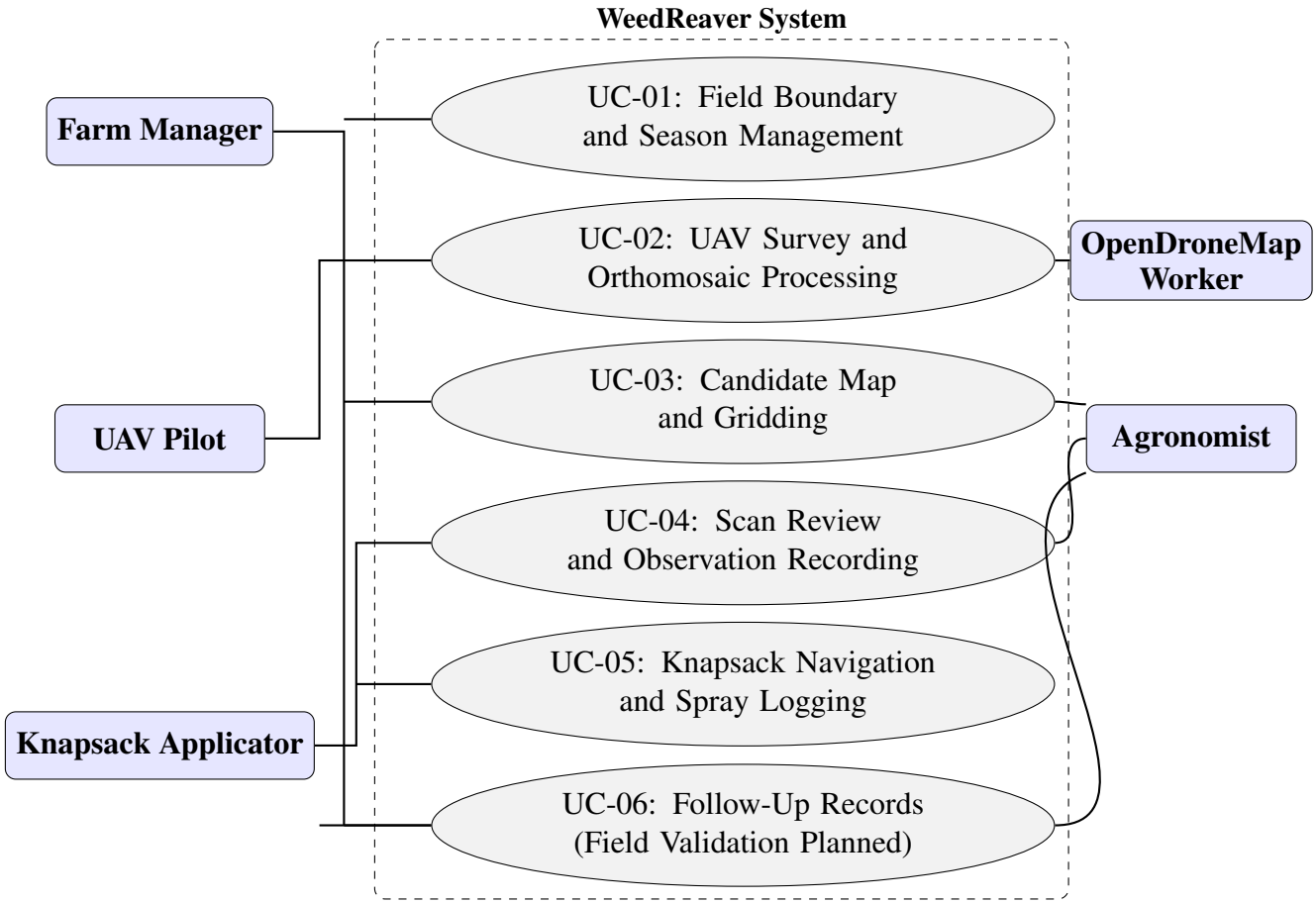


Figure 4.1: System Use Case Diagram Illustrating Stakeholder Interactions

4.3 Detailed Use Cases

Table 4.1 documents the core use case specifications governing system behavior.

Table 4.1: Formal Specification for Core System Use Cases

Use Case Element	Specification: UC-03 Aerial Segmentation and Spray-Grid Generation
Primary Actor	Farm Manager / Agronomist
Preconditions	A georeferenced orthomosaic (GeoTIFF, GSD ≤ 0.4 cm/px) has been successfully generated and bounded to a valid FieldSeason in Module C1/C2.
Trigger	User requests weed prescription generation from the GIS dashboard.
Main Success Scenario	1. System tiles the field orthomosaic into 512×512 pixel chips with a 64-pixel spatial stride buffer. 2. PyTorch deep semantic segmentation model infers pixel-level class probabilities (Crop, Grass Weed, Broadleaf Weed, Soil). 3. The selective classification layer evaluates predictive entropy against threshold τ; pixels exceeding τ are pushed to the Module C4 Ground Abstention Queue. 4. Accepted weed predictions are aggregated into a discrete 1×1 m spatial grid using projected UTM Zone 42N coordinates. 5. Each 1×1 m cell is classified as *Treated* if weed canopy coverage exceeds the 5% threshold, generating a binary spray-grid map. 6. The system computes total treated area, estimated chemical saving percentage, and exports the prescription layer.
Extensions / Exceptions	3a. If over 40% of canopy pixels trigger abstention due to severe blur or shadow, the system halts gridding, alerts the pilot to re-fly, and flags illumination anomaly. 5a. If total treated area fraction exceeds 70%, the system issues an economic advisory warning that broadcast spraying is more cost-effective than knapsack spot-spraying.
Postconditions	A georeferenced map grid and review queue are persisted in the backend and made available to supported clients through the API.

4.4 Functional Requirements

The functional requirements of WeedReaver are organized across the six core architectural modules, with each requirement assigned a unique, testable identifier.

4.4.1 Module C1: Field and Season Management

- **FR-C1.1 (Boundary Ingestion):** The system shall allow users to create and persist field parcel boundaries via manual interactive polygon drawing on a satellite basemap or via GeoJSON/KML file upload.
- **FR-C1.2 (Coordinate Transformation):** The system shall automatically project geographic coordinates (WGS84 EPSG:4326) into projected planar coordinates (UTM Zone 42N EPSG:32642) to ensure metric distance and area calculations are geometrically accurate within $\pm 0.1\%$.
- **FR-C1.3 (Field-Season Scoping):** The system shall bind all surveys, imagery, prescriptions, and treatment logs to an explicit *FieldSeason* entity defined by crop cultivar, sowing date, and parcel ID.
- **FR-C1.4 (Boundary Validation):** The system shall validate that ingested polygons are simple, non-self-intersecting, and possess an enclosed surface area between 0.25 acres and 100 acres.

4.4.2 Module C2: Orthomosaic Processing and GIS Viewer

- **FR-C2.1 (UAV Sortie Ingestion):** The system shall ingest batches of visible-spectrum RGB aerial images containing embedded EXIF GPS tags and camera intrinsic metadata.
- **FR-C2.2 (Automated Photogrammetry):** The system shall interface asynchronously with NodeODM / OpenDroneMap to execute feature matching, bundle adjustment, and orthorectification, outputting a single GeoTIFF raster.
- **FR-C2.3 (Resolution Verification):** The system shall verify that the synthesized orthomosaic achieves a Ground Sampling Distance (GSD) under 0.40 cm/pixel across 95% of the parcel area.
- **FR-C2.4 (Slippy Map Raster Tiling):** The system shall slice master GeoTIFF orthomosaics into standard 256×256 Slippy map tiles (zoom levels 16 to 22) optimized for low-latency web viewing via Leaflet.

4.4.3 Module C3: Aerial Segmentation and Spray-Grid Mapping

- **FR-C3.1 (Tiled Aerial Inference):** The system shall partition orthomosaics into 512×512 pixel tiles, execute batch segmentation inference, and stitch predicted probability rasters back into georeferenced coordinates with border blending.

- **FR-C3.2 (Four-Class Delineation):** The model shall classify each pixel into one of four mutually exclusive classes: Crop (*Triticum aestivum*), Grass Weed (*Phalaris minor* / *Avena fatua*), Broadleaf Weed, or Background (Soil/Stubble).
- **FR-C3.3 (Selective Abstention Thresholding):** The system shall compute predictive softmax entropy for each pixel; if entropy exceeds a calibrated threshold τ, the pixel classification shall be declined and flagged as *Abstained*.
- **FR-C3.4 (Spray-Grid Discretisation):** The system shall overlay a regular 1×1 m grid over the parcel and flag each cell for spray application if accepted weed pixel area within the cell exceeds 500 cm^2 (5% cell area).
- **FR-C3.5 (Prescription Metrics Generation):** The system shall compute total field area, total infested area, prescribed spray area, and theoretical herbicide chemical volume reduction relative to broadcast spraying.

4.4.4 Module C4: Ground Verification and Abstention Queue

- **FR-C4.1 (Abstention Cluster Clustering):** The system shall perform spatial clustering (DBSCAN) on abstained aerial pixels to generate discrete ground verification waypoints (latitude, longitude, radius).
- **FR-C4.2 (Mobile Queue Download):** The mobile application shall download and locally cache the abstention waypoint queue for offline field execution.
- **FR-C4.3 (Field Review):** The Android prototype shall allow an operator to record or review a scan and preserve its status. Any future automated prediction must be clearly distinguished from a human-confirmed label; no trained local weed model is currently included.
- **FR-C4.4 (Ground Observation Recording):** The mobile app shall record ground truth observations (species, count, confidence, photo voucher) and update the local database with offline sync timestamps.

4.4.5 Module C5: Prescription and Knapsack Navigation

- **FR-C5.1 (Offline Prescription Caching):** The mobile application shall cache field boundaries, spray-grid rasters, and offline map tiles locally in SQLite and device storage.
- **FR-C5.2 (Real-Time GPS Positioning):** The application shall track the applicator's real-time GNSS location, displaying operator position relative to the 1×1 m spray grid with heading orientation.

- **FR-C5.3 (High-Contrast Audio/Visual Alerts):** When the operator enters a cell designated for spray application, the mobile UI shall flash a high-contrast amber indicator and emit an audio tone.
- **FR-C5.4 (Manual Cell State Toggling):** The operator shall be able to manually toggle a cell between *Sprayed*, *Unsprayed*, and *Skipped* with a single large tap target.

4.4.6 Module C6: Treatment Record, Follow-Up and Rotation Log

- **FR-C6.1 (Application Record Logging):** The system shall record the execution metadata for each chemical application, including date, applicator ID, commercial product brand name, active chemical ingredient, and HRAC mode-of-action group.
- **FR-C6.2 (Strict Rate Non-Prescription):** The system shall strictly prohibit the automated calculation of chemical dose rates or dilution ratios, displaying a mandatory instruction to consult official registered container labels.
- **FR-C6.3 (Temporal Follow-Up Co-Registration):** The system shall ingest a follow-up UAV orthomosaic captured 14 days post-application and co-register it to the pre-spray orthomosaic using ground control points with < 1.0 cm spatial error.
- **FR-C6.4 (Efficacy Delta Computation):** The system shall compute the biological Efficacy Delta:

$$\Delta_{\text{efficacy}} = \frac{\text{WeedArea}_{\text{pre}} - \text{WeedArea}_{\text{post}}}{\text{WeedArea}_{\text{pre}}} \times 100\% \tag{4.1}$$

  and flag persistent surviving patches as potential herbicide resistance suspects.

4.5 Non-Functional Requirements

4.5.1 Reliability and Robustness

- **NFR-REL-01 (Offline Field Fault Tolerance):** The mobile application shall operate indefinitely with zero cellular or Wi-Fi network connectivity, persisting all navigational state, survey logs, and model inferences in local SQLite storage without crash or data loss.
- **NFR-REL-02 (Database and Synchronisation):** Backend transactions and Android local storage/synchronisation shall preserve record consistency across inter-

rupted connections; verify behavior with the repository's tests and a documented device run.

- **NFR-REL-03 (Photogrammetry OOM Recovery):** The backend photogrammetry pipeline shall monitor host RAM and cleanly terminate and checkpoint processing if memory usage exceeds 90%, preventing operating system panic.

4.5.2 Usability and Human Factors

- **NFR-USE-01 (Low-Literacy Mobile Interface):** All field operational screens in the mobile application shall utilize universally understood iconographic indicators, color-coded status banners, and large touch targets ($\geq 48 \times 48$ dp) operable by users with minimal formal literacy.

- **NFR-USE-02 (High-Glare Sunlight Visibility):** The mobile navigation UI shall incorporate a high-contrast outdoor mode with a minimum contrast ratio of 7:1 (WCAG AAA compliance) to ensure complete legibility under intense direct midday sunlight.

4.5.3 Performance and Execution Latency

- **NFR-PERF-01 (Future Model Performance):** After a real model and target device are selected, measure end-to-end inference latency, memory use, and thermal behavior. Do not treat demo scan timing as model performance.

- **NFR-PERF-02 (GIS Tile Rendering Performance):** The web dashboard shall load and display 256×256 Slippy map raster tiles in under 200 ms over standard local broadband connections.

- **NFR-PERF-03 (Tiled Aerial Segmentation Throughput):** The backend aerial segmentation engine shall process a 5-acre orthomosaic (approximately 2.5 gigapixels) in under 15 minutes on a workstation equipped with an NVIDIA RTX 3060 GPU.

4.5.4 Security, Privacy and Governance

- **NFR-SEC-01 (Farmer Coordinate Anonymization):** Public API endpoints and exported benchmark datasets shall strip exact cadastral parcel coordinates and owner identities, applying a random 500 m geodetic spatial jitter to protect landholder privacy.

- **NFR-SEC-02 (API Authentication):** All REST API communication between mobile clients and the backend server shall be secured via TLS 1.3 and authenticated using JSON Web Tokens (JWT).

4.6 Acceptance Criteria

Table 4.2 defines the formal, objective acceptance test criteria for verifying critical functional and non-functional requirements.

Table 4.2: System Acceptance Criteria and Objective Verification Methods

Req ID	Requirement Target	Objective Pass Condition	Verification Method
FR-C1.2	Coordinate Projection	Coordinate transformation from EPSG:4326 to EPSG:32642 introduces < 0.05 m metric error on a 100 m baseline.	Automated Pytest unit test against GDAL reference.
FR-C2.3	Orthomosaic GSD	Mean GSD across synthesized GeoTIFF is ≤ 0.40 cm/pixel, verified against calibrated ArUco ground fiducials.	Rasterio metadata inspection and ArUco edge measurement.
FR-C3.4	Spray-Grid Conversion	Continuous segmentation probabilities convert to a binary 1×1 m grid where infested cells achieve $\geq 80\%$ recall.	Confusion matrix against manual polygon ground truth.
FR-C4.3	Scan review	Record a demo scan, distinguish its predefined outcome from a model result, and verify persistence/sync.	Android client test/run record; model benchmark pending.
NFR-REL-01	Complete Offline Mode	Mobile application navigates, records 50 ground observations, and toggles 100 cells with Wi-Fi/cellular disabled without failure.	Physical device disconnected validation run.

4.7 Requirements Traceability Matrix

Table 4.3 maps user needs to requirements, repository evidence, and the status of remaining validation.

Table 4.3: Requirements Traceability Matrix and Repository Evidence

User Need	Functional Req	Design Module	Repository Evidence	Status
Boundary Management	FR-C1.1, FR-C1.2	Module C1	`backend/app/api/fields.py`; backend field tests	Implemented; test execution pending
Aerial processing	FR-C2.2, FR-C2.4	Module C2	`backend/app/pipeline/orthomosaic.py`; NodeODM tests	Adapter present; simulated default
Map and zone analysis	FR-C3.1, FR-C3.4	Module C3	`backend/app/analysis/`; grid/zone tests	Logic implemented; aerial model pending
Review and verification	FR-C3.3, FR-C4.1	Module C3 / C4	`backend/app/api/scans.py`; review records	Routes/workflow present; prediction confidence is demo data
Android prototype	FR-C4.3, NFR-REL-01	Module C4	`app/src/main/java/...`; Room and sync code	Screens/storage present; scan outcomes predefined
Route and treatment workflow	FR-C5.2, FR-C5.3	Module C5	Android route/treatment screens; backend treatment APIs	Prototype workflow; field evaluation pending
Treatment verification	FR-C6.1, FR-C6.4	Module C6	Backend treatment and verification services; service tests	Records present; demo trend is not measured efficacy

Chapter 5

System Design

5.1 System Architecture

This chapter separates the implemented repository architecture from the proposed field workflow. The current codebase provides a FastAPI backend and React/TypeScript dashboard. The processing interfaces include a NodeODM adapter, while the default orthomosaic and segmentation engines are simulated. The model-backed segmentation engine and the separate Android field client are not implemented in the audited repository. The following diagram is a target architecture, not a deployment or validation claim:

1. **Aerial data path:** UAV imagery may be processed through NodeODM when a service and suitable imagery are configured. Capture altitude and image quality are study parameters to be measured.
2. **Current software:** A FastAPI backend and React/TypeScript dashboard manage records, review, and grid/zone analysis. Synthetic processing is the default; a trained aerial model is future work.
3. **Field operation:** Kotlin/Compose Android client source is present. Its UI and local persistence/synchronization are implemented as a prototype, while scan outcomes are demo values and on-device weed inference is not implemented.

Figure 5.1 illustrates the high-level system architecture, highlighting component separation and data communication boundaries.

5.2 Architectural Components

The report uses six module labels to organize requirements. The repository's implemented boundaries are more accurately described as follows:

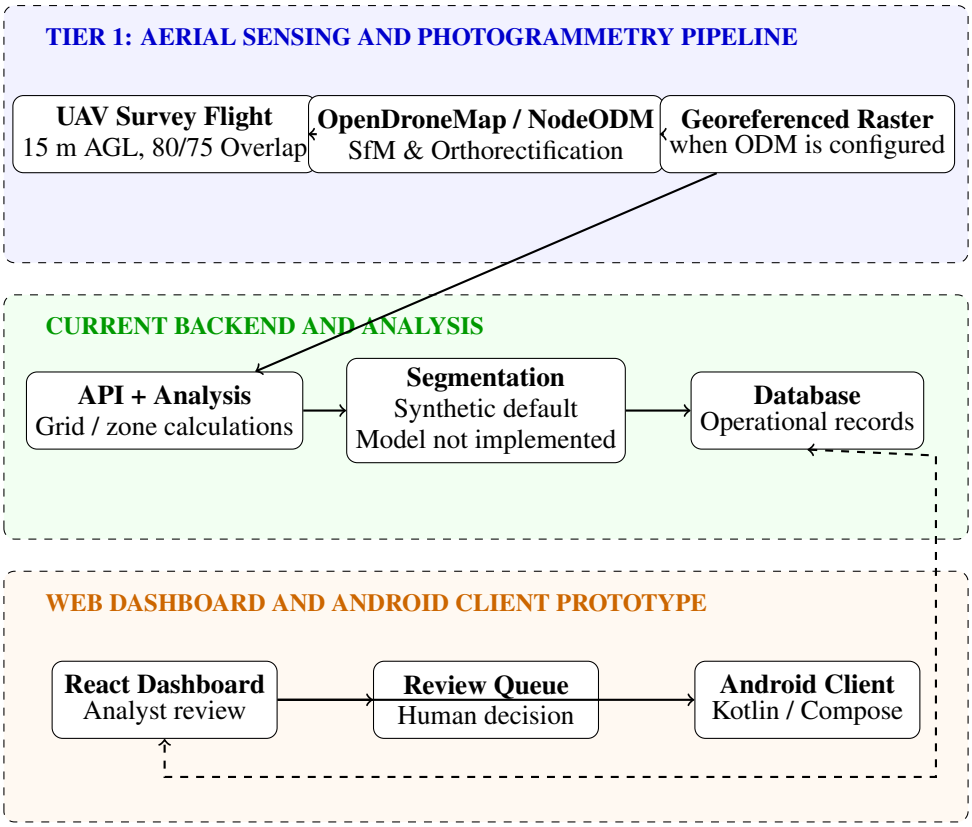


Figure 5.1: Multi-Tier System Architecture Diagram of WeedReaver

1. **Field and season records:** Backend models and API routes maintain field, crop-season, survey, and operational records.

2. **Image processing:** A NodeODM adapter is available; the simulated engine is the default and does not create an orthomosaic.

3. **Map and zone analysis:** Implemented backend and dashboard logic calculates grids, zones, and routes for stored or synthetic surfaces.

4. **Aerial segmentation:** Synthetic segmentation is used in demo mode. The model-backed adapter is not implemented, so no learned predictions or model abstention results are available.

5. **Review and user interface:** The React dashboard supports analyst review, treatment, and verification records. Kotlin/Compose Android screens are also present; their scan outcomes are predefined demo values.

6. **Follow-up records:** The software can store follow-up information; demo efficacy values are synthetic and are not agronomic measurements.

5.3 Interfaces and Dependencies

5.3.1 Backend RESTful API Contracts

The backend exposes REST/JSON routes under `/api/v1`. The paths below are examples from the repository API; transport security depends on deployment configuration:

- `POST /api/v1/fields/parse-boundary`: Parses supported boundary formats for the field workflow.
- `GET /api/v1/fields/{id}/grid?size=&threshold=`: Returns a field grid for the selected size and threshold.
- `POST /api/v1/surveys`, `POST /api/v1/surveys/{id}/images`, and `POST /api/v1/surveys/`: Create a survey, upload images, and request processing.
- `POST /api/v1/sync/push` and `GET /api/v1/sync/pull?since=`: Exchange client changes and server updates.

5.3.2 Future Mobile and Model Requirements

The following are proposed interface requirements for future mobile/model work; they are not a description of an existing runtime:

- **Input Tensor:** $[1, 224, 224, 3]$, data type `uint8` (RGB image cropped to ArUco quadrat boundary, normalized with mean 127.5 and scale 127.5); inference is paired with the selected crop-season and model version.
- **Output Tensor:** Crop-specific model output with n_{classes} probabilities over the taxonomy declared in the model manifest. The wheat model includes crop, wheat weed, and background classes; the cotton model will use locally verified cotton and weed classes. The system must not use wheat class labels or predictions for a cotton field-season.
- **Execution Target:** Select a model runtime and target Android hardware only after a model is implemented; measure latency, memory use, and offline behavior on the selected device.

5.4 Architecture Decisions and Rationale (ADR)

Table 5.1 documents eight foundational Architectural Decision Records (ADRs) detailing the alternatives considered, the chosen approach, and the technical justification.

5.5 Design Evolution

Following proposal-defense feedback, the report narrows the first evaluation to wheat, describes the dashboard workflow, and makes the data and implementation boundaries explicit:

1. **Narrowed problem:** The first evaluation concerns candidate *Phalaris minor* patches in wheat. The system does not claim all-species detection.
2. **Data boundary:** Synthetic demo surfaces are separated from real field data. Model training and accuracy claims await local imagery, reviewed labels, and a field-separated test set.
3. **User workflow:** The analyst dashboard and Android prototype are distinguished from live model inference and the requested clickable Figma prototype.

5.6 Domain Model

Figure 5.2 presents the conceptual domain model defining the core entities and relationships governing the WeedReaver platform.

5.7 Data Design and Record Model

The repository uses SQLAlchemy models with PostgreSQL in production and SQLite for local development and tests. Spatial calculations are implemented in the application; this report does not claim a PostGIS deployment. Table 5.2 is a conceptual record model for the proposed workflow, not a dump of the deployed schema.

5.8 Process and Behavior Design

The dynamic behavior of the WeedReaver platform is governed by three primary asynchronous workflows, modeled below as UML sequence interactions.

Table 5.1: Architecture Decision Records (ADR) Summary

Decision ID	Alternative Evaluated	Chosen Solution	Technical and Agronomic Justification
ADR-1: Map display	External tile dependency	Dashboard map implementation in the repository	Use the existing dashboard map workflow; any third-party imagery source and its terms should be confirmed before deployment.
ADR-2: Photogrammetry	Agisoft Metashape	OpenDroneMap / NodeODM (Docker)	Metashape requires expensive proprietary licensing. ODM is open-source, scriptable via headless REST APIs, and reproducible.
ADR-3: Evaluation split	Random tile / held-out field	Field- or spatial-block evaluation planned	Compare split strategies on local wheat data; do not assume a fixed inflation or generalization result.
ADR-4: Segmentation	Model family to be selected after data review	Not selected or implemented	Start with the wheat target and compare candidate models on field-separated data before choosing an architecture.
ADR-5: Field client	Dashboard-only / Android prototype	Kotlin/Compose prototype in repository	The source includes operator screens and local storage/sync; demo scans are predefined and user evaluation remains pending.
ADR-6: Data storage	Local demonstration database / server database	Backend persistence in repository	Offline field storage remains a separate-client requirement and is not verified here.
ADR-7: Grid unit	1 / 2 / 5 m analysis grids	Configurable grid sizes	Compare map usefulness and location uncertainty with field observations before selecting a treatment resolution.
ADR-8: Chemical Advisory	Automated Dose Calculator	HRAC Mode-of-Action Log	Calculating dosages creates severe legal liability under Pesticides Ordinance 1971. HRAC tracking guides resistance rotation safely.

Table 5.2: Conceptual Relational Record Model

Table Name	Column Name	Data Type / Constraints	Description & Spatial Indexing
fields	id name geom area_acres	UUID PRIMARY KEY VARCHAR(100) NOT NULL GEOMETRY(Polygon, 4326) NUMERIC(8,2)	Cadastral parcel boundary; spatial GIST index applied on geom.
field_seasons	id field_id crop_name cultivar sowing_date	UUID PRIMARY KEY UUID REFERENCES fields(id) VARCHAR(50) NOT NULL VARCHAR(100) DATE NOT NULL	Crop-specific season record for wheat or cotton; surveys, annotations, model evaluations, and treatment history remain linked to the crop-season.
surveys	id season_id survey_date survey_type	UUID PRIMARY KEY UUID REFERENCES field_seasons TIMESTAMP WITH TIME ZONE VARCHAR(20)	Metadata for UAV flight sortie; type is either 'pre_spray' or 'post_spray'.
spray_cells	id prescription_id cell_geom state weed_fraction	BIGSERIAL PRIMARY KEY UUID REFERENCES prescriptions GEOMETRY(Polygon, 32642) VARCHAR(20) DEFAULT 'Unsprayed' NUMERIC(5,4)	Projected 1×1 m grid cells (UTM 42N); spatial GIST index applied on cell_geom.
treatment_records	id season_id application_date product_name hrac_group	UUID PRIMARY KEY UUID REFERENCES field_seasons DATE NOT NULL VARCHAR(100) NOT NULL VARCHAR(20) NOT NULL	Logs active herbicide chemistries and modes of action for multi-season rotation audits.

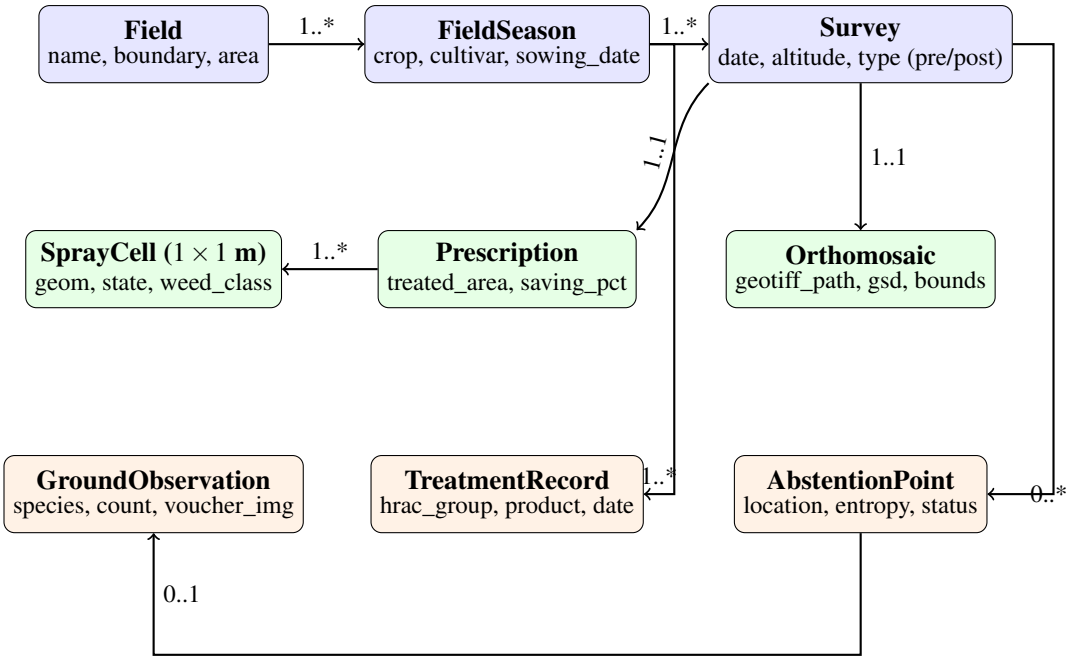


Figure 5.2: Conceptual Domain Model for WeedReaver Precision Management Loop

5.8.1 1. Survey-to-Prescription Workflow

Figure 5.3 sketches the target sequence from flight-image upload to map-grid generation. The NodeODM adapter can be configured for real imagery; segmentation remains simulated by default and its learned model adapter is not implemented.

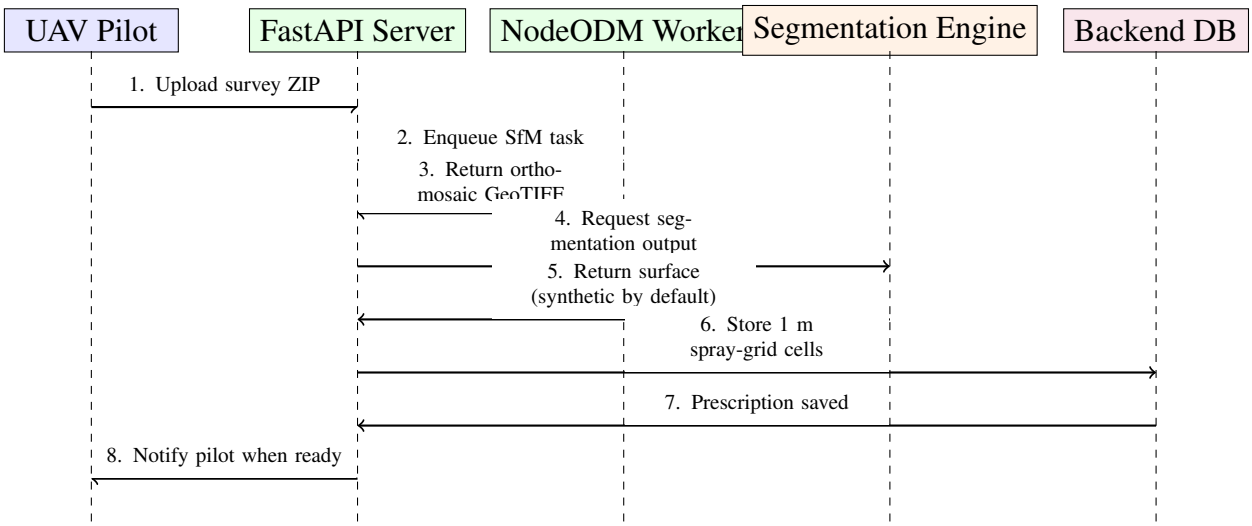


Figure 5.3: UML Sequence Diagram for Survey-to-Prescription Pipeline

5.8.2 2. Selective Abstention and Ground Verification Workflow

Figure 5.4 sketches the planned review flow. The current segmentation engine does not produce learned confidence or real abstentions; Android scan results are predefined demo outcomes.

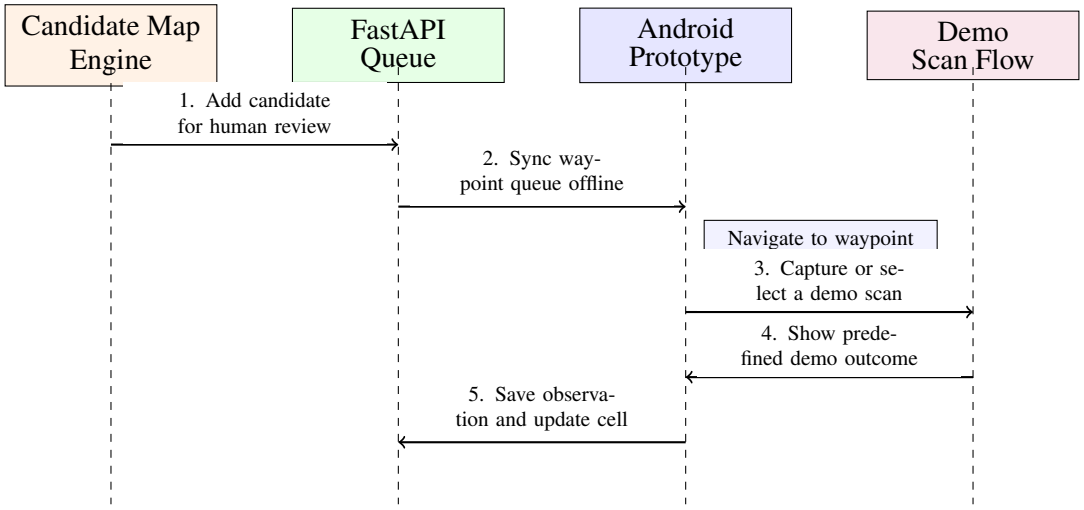


Figure 5.4: Planned Review Flow with Predefined Android Demo Outcomes

Chapter 6

Iteration 1: Design and Implementation

6.1 Repository Snapshot and Implementation Boundary

The GitHub repository was reviewed at commit `10057dcb1a68d0c0223bf213dd8d29c5e3f3477a` (10 October 2026). It contains a Python/FastAPI backend, a React/TypeScript dashboard, and Kotlin/Jetpack Compose Android client source. The demonstrated workflow and seeded data are wheat-oriented. The Android code includes offline storage/sync and user-interface workflows, but scan outcomes are predefined demo data and no on-device weed model asset is present. The README says the companion client is developed separately, which conflicts with the Android source present in this checkout; this should be reconciled in project documentation.

6.2 What the Repository Implements

The implemented baseline consists of the following connected parts:

- **Field and season records:** Backend models and API routes store fields, crop seasons, surveys, zones, treatments, verification records, and audit events. The seed dataset describes a wheat station and wheat weed labels.
- **Geospatial analysis:** Backend and dashboard code calculate infestation surfaces, grids, zones, routes, and efficacy values. The repository documents parity checks between the Python and TypeScript calculations and includes automated backend and frontend test suites.
- **Interactive analyst dashboard:** The React interface includes field, flight, weed-map, review, treatment, rotation, verification, export, device, and settings views.

The threshold control updates the displayed grid and zones, enabling an analyst to review a prescription workflow.

- **Android client source:** Kotlin/Compose screens, Room-backed local persistence, synchronization, route/treatment flows, and a scan interface are present. Scan labels and inference timings shown in the app are demo outcomes, not live model predictions.

- **Processing interfaces:** A NodeODM adapter can process uploaded imagery when configured. The default orthomosaic and aerial-segmentation engines are simulated for demos and tests. The model-backed aerial segmentation adapter raises an explicit not-implemented error.

- **Human review and records:** The backend supports scan labels, review outcomes, treatment records, and follow-up verification records. These data flows do not establish model accuracy or agronomic efficacy by themselves.

6.3 End-to-End User Workflow

The repository contains an analyst-facing web workflow. An analyst signs in, selects a field and season, reviews flight and weed-map information, adjusts the prescription threshold, inspects zones and review items, and records treatment or verification information. The Android prototype provides screens for field selection, map review, route/navigation, scan, treatment, verification, and history; the intended operator sequence is to select a field, inspect the published zones, follow a route, record work, and sync when connectivity returns. Scan results in this prototype are predefined, not classifier outputs. In a configured deployment, imagery can be sent to NodeODM for an orthomosaic. The default demo uses simulated processing data.

The coded Android screens and local persistence/synchronization are present, but the scan UI uses predefined demo outcomes and the repository contains no on-device weed model. Device-level recognition performance is therefore not evidenced. The Figma prototype requested by the panel remains a separate design deliverable; coded screens are not represented as a Figma prototype.

6.4 Crop Scope and Target Weed

The current implementation and first local evaluation are limited to Rabi wheat. The primary target is *Phalaris minor* (little-seed canary grass), a competitive grass weed that can

resemble wheat at a distance. The seeded wheat demonstration also lists *Avena ludoviciana*, *Chenopodium album*, and *Convolvulus arvensis*; these are demonstration labels, not a claim that the system reliably detects them in field imagery.

The label policy separates (1) the crop, (2) weeds selected as treatment targets, (3) other or uncertain vegetation, and (4) background. A plant is not a treatment target merely because it is green or is not wheat. Uncertain observations should be reviewed by a person; harmless or non-target vegetation should not be prescribed for treatment without agronomic confirmation.

Cotton (*Gossypium hirsutum*) remains in the report as a future crop extension, not a second validated implementation domain. Cotton has a separate Kharif calendar and weed community, including reported broadleaf, grass, and sedge species such as *Trianthema* spp., *Cynodon dactylon*, and *Cyperus rotundus* [1]. The current demo species and product catalog are wheat-focused. Cotton requires local imagery, crop-stage metadata, an agronomist-approved label set, and a separate evaluation before cotton use is claimed.

6.5 Data Plan and Research Contribution

The repository's demonstration infestation surfaces and Android scan outcomes are synthetic or predefined; they are useful for exercising map, grid, zone, sync, and treatment-record behavior, but cannot serve as training or validation data. The repository includes no labeled Punjab aerial dataset and no trained aerial segmentation model. Public datasets may be used to prototype the training pipeline, but any such result must name its source and must not be described as local Punjab performance.

The next research step is to collect georeferenced RGB imagery from wheat fields at recorded growth stages and pair it with field observations. The data protocol will record field, date, crop stage, camera/flight metadata, label source, annotator, and weed/crop/other/uncertain class. An agronomist will confirm target labels. Training and evaluation partitions must be separated by field or spatial block so adjacent image tiles do not appear in both. Cotton data, if later acquired, will be stored and evaluated separately.

The defensible contribution at this checkpoint is an implemented, human-reviewable geospatial workflow for survey records, grid/zone generation, treatment records, and follow-up data, plus a testable plan for spatially independent evaluation. Weed detection quality, treatment savings, efficacy, and crop-yield effects remain research questions; no achieved numerical results are claimed here.

6.6 FYP-2 Work Plan

1. Complete the wheat data protocol and confirm field access, flight permissions, and agronomist review before capture.
2. Acquire a small, documented wheat dataset centred on *Phalaris minor* and wheat, with other vegetation and uncertain cases retained as separate labels.
3. Implement and evaluate an aerial segmentation model only after the data and label quality are adequate; report field-separated metrics and failure examples.
4. Compare prescription maps with field observations and record whether zones are useful to a human operator. Do not infer herbicide savings or efficacy from synthetic demo surfaces.
5. Treat cotton as a separately scoped extension. Proceed only after a Kharif site, local labels, and a feasible schedule are confirmed.

6.7 Technology Stack

Table 6.1: Technology visible in the checked repository and its current role

Layer	**Technology**	**Current repository role**
Backend	Python, FastAPI, SQLAlchemy, Alembic	API, domain rules, persistence, processing jobs, and tests.
Database	PostgreSQL in production; SQLite for local work/tests	Stores user, field, season, survey, zone, treatment, review, and sync records.
Dashboard	React, TypeScript, Vite	Analyst-facing interactive workflow and map presentation.
Photogrammetry	OpenDroneMap through NodeODM adapter	Available when configured; simulated engine is the default.
Aerial segmentation	Pluggable engine interface	Synthetic engine is the default; model-backed engine remains unimplemented.
Field application	Separate Android application described by project documentation	Not included in the audited repository; no on-device model or latency result is claimed.

Chapter 7

Testing and Early Validation

7.1 Current Evidence Boundary

The repository contains automated backend tests and frontend component/data tests. Its README describes these suites and gives local commands for running them. This report does not claim a specific test count, coverage percentage, physical-device result, or field validation unless an execution record is available with the deliverable. A test file or demo screen is evidence of intended behavior, not proof of agronomic performance.

7.2 What Can Be Tested Now

The current software supports engineering tests for authentication and roles, field and season records, geometry parsing, grid and zone calculations, routes, treatment and review records, exports, and offline synchronization contracts. The React dashboard also contains tests for API clients, adapters, geometry, and the demonstration infestation model. These tests use fixtures and synthetic data; passing them would establish software behavior for those inputs only.

The NodeODM adapter can be tested against a configured NodeODM service and suitable image set. The default orthomosaic engine is simulated and produces no raster. Likewise, the default aerial-segmentation engine creates synthetic infestation surfaces, while the model-backed adapter is not implemented. The repository therefore cannot support a model-accuracy claim or a real efficacy estimate from its default demo.

7.3 Planned Validation

1. **Engineering checks:** Run the backend and frontend suites from the repository instructions; preserve the command, environment, commit, and complete output with the report.

2. **Image-processing checks:** Once field imagery is available, verify orthomosaic georeferencing, ground sampling distance, and visible weed patch alignment against surveyed points.

3. **Label checks:** Have an agronomist review a documented sample of crop, target weed, other vegetation, and uncertain labels. Report agreement and resolve disagreements before model training.

4. **Model checks:** Compare the wheat model on field-separated data, report class-wise precision/recall and confusion examples, and include abstention coverage only after a model with confidence output is implemented.

5. **Workflow checks:** Ask intended users to complete field selection, map review, zone decision, treatment recording, and follow-up tasks. Record completion errors and usability findings. Validate any separate Android client independently.

6. **Agronomic checks:** Measure treatment response against field observations and an appropriate comparison plot before reporting efficacy or input savings. Keep cotton results separate and out of scope until local Kharif data exist.

7.4 Test Case Register

7.5 Validation Limitations

No locally collected Punjab aerial dataset, trained aerial model, independently run field experiment, or validated wheat or cotton efficacy result was present in the repository snapshot audited for this report. Demo patch values, threshold choices, and treatment trends must not be interpreted as measured field outcomes. These are explicit limitations and determine the next validation work.

Table 7.1: Planned test cases and the evidence required to close them

ID	Area	Check	Status
TC-01	Field geometry	Accept valid boundary files and reject malformed or self-intersecting geometry.	In repository; run evidence pending
TC-02	Grid and zones	Compare grid, zone, and route outputs with fixed reference fixtures.	In repository; run evidence pending
TC-03	Survey processing	Verify image upload and NodeODM output metadata with a real configured service.	Pending real processing evidence
TC-04	Simulated processing	Clearly mark synthetic map and follow-up values as demo output.	Current demo behavior
TC-05	Crop and weed labels	Confirm wheat, target weed, other vegetation, and uncertain labels with an agronomist.	Pending field data
TC-06	Wheat model	Evaluate on field-separated imagery and report class-wise metrics and errors.	Pending model and data
TC-07	Cotton extension	Repeat data and evaluation protocol on local Kharif cotton before reporting results.	Future extension
TC-08	User workflow	Observe analyst dashboard tasks and separately validate the field-operator client.	Usability evidence pending

Chapter 8

Project Management and Development Evidence

8.1 Repository Audit

The implementation statements in this chapter are based on a read-only review of the public GitHub repository at commit `10057dcb1a68d0c0223bf213dd8d29c5e3f3477a`, dated 10 October 2026. The report source is maintained separately. No repository files were edited and no commit or push was made as part of this report update.

8.2 Work Completed in the Repository

The verifiable implementation work is the backend and web application described above. The web dashboard exposes analyst workflows for fields, flights, a weed map, review, treatment, rotation, verification, exports, devices, and settings. Backend analysis derives map grids, treatment zones, routes, and verification summaries from stored or demonstration surfaces. The system also contains configurable processing-engine interfaces and a concrete NodeODM adapter.

The checked-in default configuration selects the simulated orthomosaic engine and simulated segmentation engine. The simulation can produce believable demonstration values, including synthetic follow-up trends, but these values are not observations from a farm. The code includes a wheat-focused seed catalog with *Phalaris minor*, *Avena ludoviciana*, *Chenopodium album*, *Convolvulus arvensis*, and wheat as a crop label. It includes wheat product records; it does not establish a cotton label set or locally validated cotton workflow.

Table 8.1: Repository components and evidence boundary

Component	Present in repository	What the evidence supports
Backend	FastAPI application, SQLAlchemy models, migrations, API routes, services, processing interfaces, and tests.	A substantial software baseline for user, field, season, survey, map-analysis, treatment, verification, export, and sync workflows.
Web dashboard	React and TypeScript application with map and workflow views, API client, data adapters, and tests.	An interactive analyst-facing interface. It is not evidence of a completed Figma prototype or field-user validation.
Photogrammetry	NodeODM/OpenDroneMap adapter and a simulated default engine.	Real reconstruction is possible when configured with a NodeODM service and imagery; demo mode does not produce an orthomosaic.
Aerial segmentation	Segmentation interface and synthetic default engine; model adapter explicitly unimplemented.	Synthetic grid/zone behavior can be demonstrated. No trained aerial model or weed-detection performance is evidenced.
Field data	Seeded wheat demonstration station, synthetic patch surfaces, and demonstration observations.	Useful for exercising screens and backend logic, but not a local training or validation dataset.
Android client	Kotlin/Jetpack Compose source with Room storage, synchronization, and field-workflow screens. README says the companion app is separate, creating a documentation mismatch.	UI and persistence/sync code are present. Scan outcomes are predefined demo data and no on-device weed model asset was found; model accuracy and latency are not evidenced.

8.3 Data and Experiment Work Remaining

1. Confirm a wheat field, permissions, capture window, and agronomist involvement before collecting imagery.

2. Record raw imagery provenance, field boundaries, crop and growth stage, flight metadata, survey date, and ground observations.

3. Build an annotation guide that distinguishes wheat, *Phalaris minor*, other confirmed weeds, non-target vegetation, and uncertain cases. Retain provenance and reviewer decisions.

4. Implement and version a real aerial model, then evaluate it on fields excluded from training. Report failures and crop-specific metrics, not only an aggregate score.

5. Test map review and treatment recording with intended users. Keep claims about spray savings and efficacy pending a controlled comparison and field measurements.

6. Scope cotton only after wheat implementation boundaries are stable. Cotton requires local Kharif imagery, separate labels, and independent model and workflow evaluation.

8.4 Repository History and Contribution Records

The audited checkout was a shallow read-only snapshot, so it does not provide a complete commit history from which to substantiate individual code authorship, branch protections, pull-request approvals, or release tags. Earlier draft commit identifiers and Git-Flow descriptions have therefore been removed from this report. Individual contribution statements should be completed from verified team records before submission.

8.5 Tools and Dependencies

The repository identifies Python 3.11+, FastAPI, SQLAlchemy, Alembic, NumPy, PostgreSQL (with SQLite for local work), React 19, TypeScript, Vite, Kotlin/Jetpack Compose, Room, Retrofit, and NodeODM/OpenDroneMap as project technologies. Exact installed versions and successful execution are environment-dependent. This report does not claim use of Flutter, LiteRT, PostGIS, DeepLabv3+, or training hardware as completed repository deliverables.

8.6 Risks and Limitations

- **No local aerial labels or implemented model:** The principal research result depends on gathering suitable wheat imagery, annotation, implementing the aerial model, and evaluating it independently.
- **Synthetic demo can be mistaken for evidence:** Screens and generated infestation/effectiveness values must be visibly identified as demonstration data.
- **Processing resources:** Real photogrammetry requires a configured NodeODM host with sufficient storage and memory; the actual processing envelope must be measured on the chosen hardware and image set.
- **Crop scope:** The current evidence is wheat-focused. Cotton cannot be claimed as supported for agronomic decisions until its crop-specific catalog, local data, and evaluation are completed.
- **Field-user workflow:** Android client source is present, but the scan flow uses predefined demo outcomes. Field-user study evidence and real on-device recognition remain outstanding.

8.7 Defense Feedback Addressed in the Report

The report now narrows the first study to wheat and names *Phalaris minor* as its primary target. It distinguishes crop plants, treatment-target weeds, other vegetation, and uncertain cases, and it states that unverified plants must not be treated as weeds. It identifies the repository's synthetic data and unimplemented model boundary, provides a user workflow for the analyst dashboard, separates planned field-operator and cotton work from completed code, and replaces unsupported implementation and test claims with a staged data and validation plan.

The panel's request for a clickable Figma prototype remains an external deliverable. This report describes the interactive dashboard in the repository but does not claim that a Figma prototype has been built or tested.

Chapter 9

Remaining FYP Plan

9.1 Scope for the Next Phase

The next phase should first complete a bounded wheat study, using Punjab Rabi wheat and candidate *Phalaris minor* patches. Cotton (*Gossypium hirsutum*) is retained as a later extension, with a separate Kharif data plan. The project should not claim a cotton detector or transfer wheat metrics to cotton without local cotton data.

The FYP-2 plan is conditional on field access, permissions, equipment, agronomist availability, and data quality. Candidate sites such as NARC Islamabad or BARI Chakwal require confirmation; this report does not state that access has been granted.

9.2 Work Packages

9.3 Data Collection and Annotation

For each wheat image set, record field identifier, date and time, crop and growth stage, camera and flight settings, georeferencing source, and any preprocessing. Ground observations should include a location or plot reference, species label, evidence photograph where feasible, annotator, reviewer, and confidence. Labels should distinguish wheat, confirmed *P. minor*, other confirmed weeds, non-target vegetation, and uncertain/background.

Do not use the seeded demo surfaces as training or validation labels. Before training, define a field- or spatial-block split so adjacent image tiles from the same location cannot appear in both training and evaluation. Report the split manifest and data limitations with every result. A cotton dataset, if collected, must have its own species guide, crop-stage

Table 9.1: Remaining work packages and evidence

Work package	**Tasks**	**Exit evidence**
W1: Data readiness	Confirm field access, permissions, capture settings, weed labels, and review process.	Approved protocol and metadata sheet.
W2: Wheat pilot collection	Capture georeferenced wheat imagery and ground observations at documented crop stages.	Traceable image set and reviewed labels.
W3: Aerial model	Implement a model-backed engine and evaluate with a field-separated split.	Versioned model, class-wise metrics, and error examples.
W4: Workflow validation	Test analyst map review and Android prototype tasks with intended users; finish the clickable Figma prototype.	Observed task outcomes and revised prototype.
W5: Field treatment study	Compare mapped zones and follow-up observations under an agreed, safe field protocol.	Measured results with limitations.
W6: Cotton extension gate	Confirm Kharif site, labels, time, and agronomic review before collection or model work.	Written go/no-go decision and, if feasible, separate cotton data.

metadata, and evaluation split.

9.4 Model and Workflow Evaluation

First establish that the image and label data are suitable. Then compare a small number of segmentation baselines using class-wise precision, recall, and intersection-over-union, plus representative false-positive and false-negative examples. If a confidence or abstention mechanism is implemented, report both coverage and performance on accepted predictions. No numerical target should be fixed until the team confirms class balance, evaluation design, and intended use.

Evaluate dashboard tasks such as selecting a field, finding a survey, changing the threshold, reviewing a zone, and recording a decision. Evaluate the Android prototype separately, clearly distinguishing offline storage/synchronization from model-backed scanning. Use the Figma prototype to demonstrate clickable navigation and collect feedback before relying on it as a usability result.

9.5 Field and Agronomic Validation

Field trials require site permission, landholder consent, flight authorization, and a protocol approved by the supervisor and relevant agronomic collaborators. Compare map recommendations with field observations before any treatment. If treatment response is studied, select a suitable comparison design and record treatment details from approved labels; do not have the software calculate application dosage.

Report follow-up observations using the crop's stage and treatment context. Do not treat simulated demo trends as efficacy. Savings, resistance, and yield effects require a measured comparison and adequate replication; they are not guaranteed project outcomes.

9.6 Indicative Schedule

The schedule below is a planning sequence rather than a confirmed site or flight booking. Dates should be updated after the team confirms access and the academic calendar.

Table 9.2: Indicative FYP-2 sequence

Period	**Planned activity**	**Gate**
Initial phase	Confirm scope, data protocol, site, permissions, resource plan, and Figma workflow.	No capture until access and safety are confirmed.
Field window	Collect and review wheat imagery and ground labels.	Continue only if data quality and labels are adequate.
Model phase	Implement baseline and run field-separated evaluation.	Report limitations and failures alongside metrics.
Workflow phase	Evaluate dashboard and Android prototype with intended users.	Revise flow and complete clickable Figma prototype.
Validation phase	Conduct follow-up observation or treatment comparison if time and approvals allow.	No efficacy claim without measured comparison.
Cotton gate	Decide whether a separate Kharif pilot is feasible.	Keep cotton out of results unless local data are collected.

9.7 Project Responsibilities

Assign each work package to a named team member and record the assignment in the project log. The audited repository snapshot does not provide enough complete history to substantiate the earlier report's detailed individual code ownership; contribution statements should be updated from team and version-control records before submission.

Chapter 10

Conclusion

10.1 Project Focus

WeedReaver is a software platform for survey-guided weed mapping and treatment records. The immediate study is limited to Rabi wheat in Punjab, with *Phalaris minor* as the primary target weed. The label policy separates crop plants, confirmed target weeds, other vegetation, and uncertain observations. Cotton (*Gossypium hirsutum*) is retained as a future extension requiring its own Kharif data and validation. The report does not transfer wheat weed, efficacy, or model claims to cotton.

10.2 Current Implementation

The repository contains a FastAPI backend and an interactive React/TypeScript dashboard. Backend and frontend code support field and season records, geospatial grid and zone calculations, flight and processing records, review, treatment, verification, exports, and related operations. The repository also contains a NodeODM adapter.

Important boundaries remain: demo field and infestation data are synthetic; simulated photogrammetry produces no raster; synthetic segmentation is the default; and the model-backed segmentation adapter is not implemented. Kotlin/Compose Android client source is present, but scan outcomes are predefined demo values and no on-device model asset was found. The README's claim that the field app is separate conflicts with this source tree. Accordingly, no aerial model accuracy, device inference latency, local treatment efficacy, chemical savings, or yield improvement is reported as achieved.

10.3 Research Contribution and Next Evidence

The software contribution is an interactive, geospatial workflow that can connect field and survey records to reviewable grids and treatment zones. Its research contribution remains to be tested. The next study needs locally collected wheat imagery, traceable agronomist-reviewed labels, evaluation separated by field or spatial block, a real model implementation, user evaluation of the dashboard and operator workflow, and field measurements before reporting agronomic impact. Cotton work must be a separate study if a suitable Kharif site and schedule become available.

10.4 Response to Proposal Defense Feedback

The report narrows the first evaluation to a defined crop and target weed, avoids all-species detection claims, clarifies target and non-target labels, and distinguishes repository code from simulated behavior and planned work. It describes the dashboard and Android client source while identifying the demo scan behavior, README/source mismatch, and absence of a completed clickable Figma prototype. Diagrams should not imply that a planned or simulated component is field-validated.

10.5 Conclusion

The current repository is a functional software and demonstration baseline, not a validated agricultural detector. Its value will be established only after real wheat field data, model evaluation, and user and agronomic validation are complete. Cotton remains visible in the broader project direction while its evidence and implementation stay separate from the wheat study.

Appendix A

Proposal Defense Feedback and Action Register

A.1 Purpose

This register maps the panel's recommendations to changes made in this report and to work that remains outside the report. Status refers to the documented report or repository state, not to an assumed completion of engineering or presentation work.

A.2 Actions

Table A.1: Response to proposal defense feedback

Panel recommendation	**Report response / remaining work**	**Status**	**Evidence**
Narrow the problem and avoid claims such as detecting every weed.	The first evaluation is bounded to Punjab wheat and candidate *Phalaris minor* patches. Other vegetation and uncertain cases are kept separate.	Updated in report	Chapters 1, 6, and 9.

Table A.1: Response to proposal defense feedback

Panel recommendation	**Report response / remaining work**	**Status**	**Evidence**
Explain the harmful and harmless weed targets.	The label policy distinguishes crop, confirmed treatment-target weed, other vegetation, and uncertain observation. Species labels require agronomist confirmation.	Updated; field review pending	Chapters 1 and 6.
Strengthen the background and state the research contribution.	The proposed contribution is stated as a testable workflow contribution; data collection, model performance, and agronomic value remain to be established.	Updated; evidence pending	Chapters 2, 6, and 9.
Clarify data and resource requirements.	Synthetic demo surfaces are explicitly excluded from training and validation. The wheat data protocol and real NodeODM resource measurement are listed as next steps.	Updated; execution pending	Chapters 6, 8, 9, and 10.
Show the front-end workflow and clickable Figma prototype.	The analyst journey is described using the interactive dashboard present in the repository. The panel-requested Figma prototype and validation of the separate field-operator client remain outstanding.	Partially addressed	Chapters 5, 6, and 9.
Make diagrams more visual and reduce text; improve presentation delivery and time management.	Report diagrams and captions are framed around the user workflow and implementation boundary. Slide density, spoken delivery, and presentation timing require a separate rehearsal and slide revision.	Report updated; presentation work pending	Chapters 5 and 6.

Table A.1: Response to proposal defense feedback

Panel recommendation	**Report response / remaining work**	**Status**	**Evidence**
Add cotton while keeping the scope realistic.	Cotton is retained as a separate future extension with distinct Kharif data and evaluation. The current repository and first study remain wheat-focused.	Scope clarified	Chapters 1, 6, and 10.
Explain how the solution will work and what has actually been built.	The report separates the FastAPI backend and React dashboard from simulated default processing, the unimplemented model adapter, and the separate Android client.	Updated in report	Chapters 5, 6, 8, and 9.

Appendix B

Detailed Requirements Specification

B.1 Extended Software Requirements Specification (SRS)

This appendix documents the complete, formal Software Requirements Specification (SRS) for WeedReaver, supplementing the core requirements presented in Chapter 4.

B.2 Detailed Use Case Specifications

B.2.1 UC-01: Cadastral Field Boundary and Season Setup

- **Primary Actor:** Farm Manager / Landowner.
- **Preconditions:** User is authenticated on the Web GIS Dashboard.
- **Trigger:** User selects "Add New Field" from the dashboard.
- **Main Success Scenario:**
    1. User navigates the Leaflet map to the target farm parcel in Punjab.
    2. User activates the polygon drawing tool and clicks boundary vertices around the parcel, or uploads a standard GeoJSON boundary file.
    3. System validates that the polygon is simple, closed, non-self-intersecting, and has an area between 0.25 and 100 acres.
    4. System re-projects vertices from WGS84 (EPSG:4326) to UTM Zone 42N (EPSG:32642) and computes the exact surface area in acres and hectares.
    5. User enters season metadata: Crop Name (Wheat or Cotton), the crop-appropriate cultivar, planting date, and crop-specific growth-stage observations. Cotton

is recorded as *Gossypium hirsutum*; wheat is recorded as *Triticum aestivum*. Weed labels, model versions, survey dates, and advisory records are associated with the selected crop-season so evidence is not mixed across crops.

6. System creates a new `FieldSeason` record in the backend database and returns a unique identifier.

- **Postconditions:** Parcel geometry and season details are persisted by the backend and available through its API.

B.2.2 UC-02: UAV Survey Ingestion and Orthomosaic Processing

- **Primary Actor:** UAV Survey Pilot.
- **Preconditions:** Autonomous UAV survey sortie has completed; raw geotagged RGB frames are stored on local media.
- **Trigger:** Pilot uploads image directory via the Survey Ingestion Portal.
- **Main Success Scenario:**

1. Pilot selects the target `FieldSeason` and uploads a ZIP archive containing 100–150 raw survey frames.
2. System parses EXIF metadata, verifying altitude (15 m $\pm$2 m AGL), nadir camera gimbal pitch (-90°), and spatial overlap ($> 75\%$).
3. System dispatches an asynchronous background task to the NodeODM container.
4. NodeODM executes OpenSfM feature extraction, bundle adjustment, dense matching, and orthorectification.
5. System verifies that the generated GeoTIFF achieves Ground Sampling Distance ≤ 0.40 cm/px.
6. System triggers `ortho_tiler.py` to generate 256×256 Slippy map tiles for web rendering.

- **Postconditions:** A georeferenced orthomosaic raster is stored on the filesystem and registered in the database.

B.2.3 UC-04: Ground Verification with Selective Abstention Queue

- **Primary Actor:** Agronomist / Field Applicator.

- **Preconditions:** Aerial segmentation has executed; abstention pixels ($H(p) > \tau$) have been clustered into waypoints; mobile client has synchronized offline cache.
- **Trigger:** Operator launches "Ground Verification" on the Android mobile application.
- **Main Success Scenario:**
  1. Mobile client displays a prioritized list of abstention waypoints sorted by proximity.
  2. Operator selects a waypoint; the moving-map GPS screen guides the operator to within 1.0 meter of the target cluster.
  3. Operator places the 1×1 m PVC quadrat frame over the vegetation.
  4. Operator aligns the phone camera over the quadrat. OpenCV detects the four ArUco corner fiducials and crops the perspective-rectified quadrat interior.
  5. The current prototype records a scan outcome for review. Any future model-backed prediction must record its model version and crop context; it must not be described as a verified local classification until model evaluation is complete.
  6. Operator reviews the prediction, confirms or edits the species label, records weed stem count, and saves the observation.
- **Postconditions:** Ground observation is persisted in mobile SQLite with timestamp, GPS fix, and photo voucher.

B.2.4 UC-05: Knapsack Prescription Navigation and Spray Logging

- **Primary Actor:** Knapsack Field Applicator.
- **Preconditions:** Spray-grid matrix (1×1 m) is cached locally in the mobile SQLite database.
- **Trigger:** Applicator enters the field parcel and selects "Start Spraying".
- **Main Success Scenario:**
  1. Mobile app displays the high-contrast moving-map grid centered on the operator's current GPS position.
  2. The navigation engine computes and displays the optimal serpentine walking path across prescribed spray cells.
  3. As the operator walks into a cell flagged as *Prescribed*, the UI flashes an amber alert banner and emits an audible beep.

4. Applicator activates the manual knapsack spray wand, treating the 1×1 m area.
5. Applicator taps the large onscreen “Mark Sprayed” button; the cell state transitions from *Unsprayed* to *Sprayed* in SQLite.
6. When all prescribed cells are treated, the app displays total treated area, chemical volume applied, and session duration.

- **Postconditions:** Real-time spray progress is logged locally and queued for synchronization upon network reconnection.

B.3 Extended Non-Functional Requirements Catalog

Table B.1 documents the formal non-functional requirements, specific quantitative metrics, and testing verification techniques.

Table B.1: Comprehensive Non-Functional Requirements and Verification Metrics

Requirement ID	**Category**	**Quantitative Metric / Target**	**Verification Method**
NFR-REL-01	Offline Reliability	100% uptime with cellular data and Wi-Fi disabled across 45-minute continuous run.	Physical disconnected field testing.
NFR-PERF-01	Inference Latency	End-to-end forward pass $<$ 150 ms on ARM Cortex-A76 CPU or NNAPI delegate.	100-run automated latency benchmark driver.
NFR-PERF-02	Tile Server Speed	256×256 Slippy map raster tile returned in $<$ 200 ms under 20 concurrent requests.	Apache Bench (ab) load testing harness.
NFR-USE-01	Accessibility (DEI)	Review readable text, contrast, and touch targets with intended field users and adapt based on findings.	Android prototype accessibility review and user study.
NFR-SEC-01	Data Privacy	Field GPS coordinates jittered by $\pm$500 m in public research datasets; zero farmer PII exposed.	Exported GeoJSON schema auditing.
NFR-SEC-02	API Security	REST endpoints protected by TLS 1.3 and cryptographically signed JWT bearer tokens.	OWASP ZAP automated vulnerability scan.

Appendix C

Test Evidence and Validation Checklist

C.1 Repository Test Structure

The repository includes backend tests under `backend/tests` and frontend tests in the web application. The project README documents commands for running them. This appendix records the test areas and evidence needed; it does not reproduce an execution transcript or claim a pass result because a dated run log was not available in the audited repository snapshot.

C.2 Evidence to Record for Each Run

For every submitted test result, retain the repository commit, command, runtime and dependency versions, operating system, test summary, failures, and coverage output when applicable. Keep unit-test evidence separate from field-data, model, user-study, and agronomic evidence.

C.3 Validation Checklist

Table C.1: Validation checklist for the current software and the planned wheat study

ID	Area	Evidence required	Status
1	Backend and API	Dated output from repository backend test command, including failures.	Run pending

Table C.1: Validation checklist for the current software and the planned wheat study

ID	Area	Evidence required	Status
2	Dashboard	Dated output from web test and production build commands.	Run pending
3	Geometry and map analysis	Fixed fixtures with expected grid, zone, and route results; confirm Python/web parity.	In repository; run pending
4	Real photogrammetry	NodeODM task record, image set provenance, georeferenced orthophoto metadata, and resource use.	Configure and measure
5	Wheat labels	Field, date, growth stage, annotator, agronomist review, and label agreement record.	Field data pending
6	Aerial model	Model/version, field-separated split manifest, class-wise metrics, error examples, and abstention coverage.	Model pending
7	Dashboard usability	Observed analyst task completion, errors, and feedback.	Study pending
8	Mobile workflow	Separate client version, device/environment, sync behavior, and task observations.	Outside repository audit
9	Agronomic impact	Treatment protocol, untreated/comparison plots, follow-up observations, and analysis.	Field trial pending
10	Cotton extension	Independent Kharif data and evaluation; do not infer from wheat metrics.	Future scope

C.4 Interpretation

Passing software tests would show that tested code paths behave as specified for the tested inputs. It would not show that weeds were correctly detected in field imagery, that a grower can use the workflow, or that treatment reduced weeds or increased yield. Those

claims require the corresponding data and study evidence.

Appendix D

Repository Structure and Audit Reference

D.1 Repository Reference

The report was compared with the public repository at `https://github.com/asadsheikh91/WeedReaver`. The audited HEAD was `10057dcb1a68d0c0223bf213dd8d29c5e3f3477a` (10 October 2026). The checkout used for review was read-only; no repository files were changed.

D.2 Top-Level Structure

```
backend/    FastAPI API, domain logic, models, migrations, tests
web/        React and TypeScript dashboard, map and workflow screens, tests
app/        Android project files present in the checkout
docs/       architecture and design assets, OpenDroneMap sample output
```

The README distinguishes the offline Android field application from this repository and states that the paired application is developed separately. The presence of Android project files is therefore not treated as evidence that the described field application or its model workflow is complete.

D.3 Processing and Data Notes

The backend defaults to simulated orthomosaic and segmentation engines. The NodeODM adapter is configurable for real processing. The segmentation model adapter is a placeholder that reports that the model is not implemented. The seed data describe a wheat

demonstration station with synthetic infestation surfaces and wheat-focused species and product records. These records support software demonstrations only; they are not local agronomic observations or model-training labels.

Appendix E

Additional Design Diagrams

E.1 Overview of Supplementary Architectural Diagrams

This appendix provides conceptual design diagrams, including the record Entity-Relationship Diagram (ERD) and a Level 1 Data Flow Diagram (DFD). The diagrams describe the intended workflow and are not deployment evidence.

E.2 Entity-Relationship Diagram (ERD)

Figure E.1 illustrates the conceptual records for fields, seasons, surveys, map zones, treatments, and review. The repository uses SQLAlchemy-backed PostgreSQL/SQLite persistence; the figure is not a physical PostGIS schema.

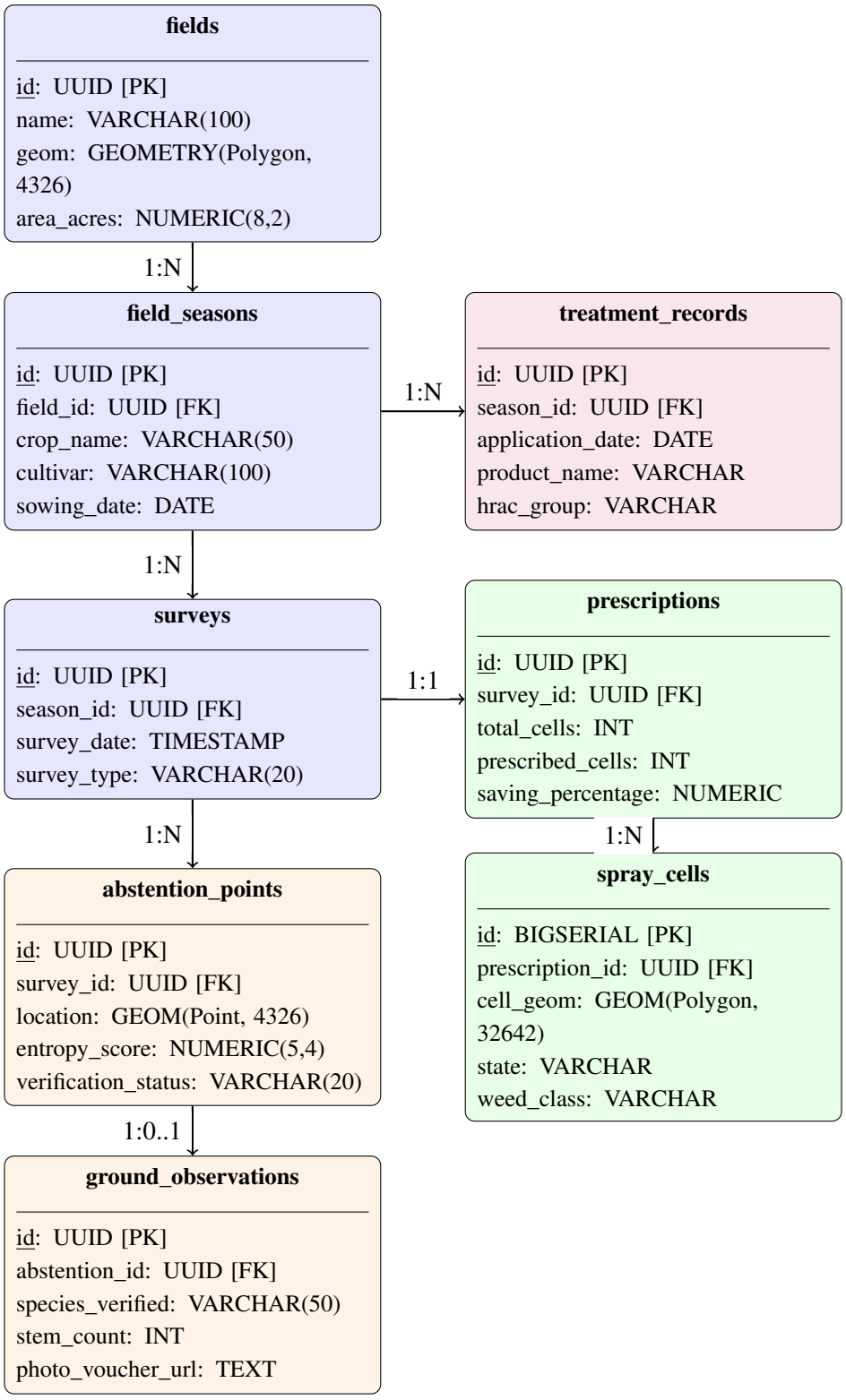


Figure E.1: Conceptual Record Relationships for the WeedReaver Workflow

E.3 Data Flow Diagram (DFD Level 1)

Figure E.2 summarizes the intended record flow. Real imagery processing is configurable; candidate-map segmentation is synthetic by default, and follow-up values require real field observations.

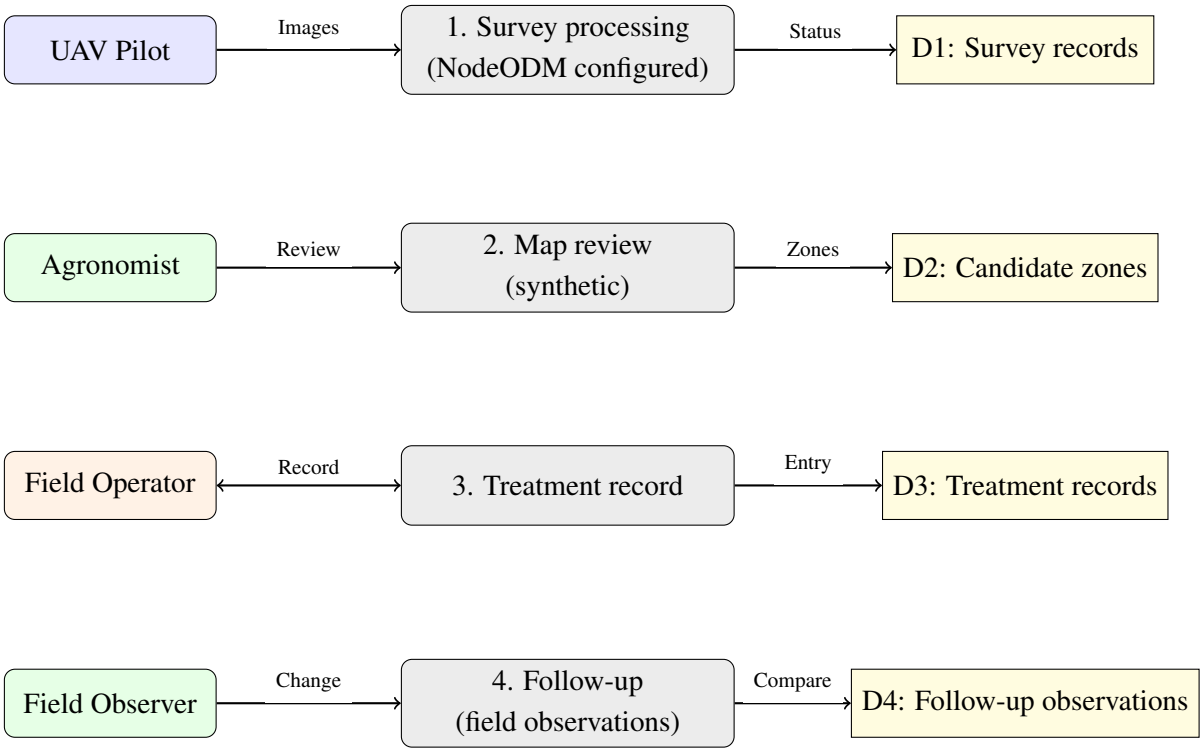


Figure E.2: Level 1 Data Flow Diagram for WeedReaver Precision Management Loop

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